

Everything You Need To Know About MAT436

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A Important Intuition 72

These notes compromise my notes on Banach spaces, Hilbert spaces, and the Spectral theory.

A blanket assumption will be that all vector-spaces will be over fields of characteristic 0. Finite characteristic fields are a different thing entirely which I'm not well aware of yet, but from what I've seen in even galois theory, it acts quite differently. As a consequence of this, all fields k we will work with will be between $\mathbb{Q} \subseteq k \subseteq \mathbb{C}$ (up to ring isomorphism). In particular, we have the nice fact that for any vector-space V , we have a nice dual V^* which will map to a space k which (after inheriting the subspace topology of $\mathbb{C}$) has all the nice topological notion we may desire.

The first thing to do is to introduce a new way of looking at vector-spaces beyond the universal property of free modules. Recall that the space $L^p(\mathbb{R}^n)$ (for $n \in \mathbb{N}_{>0}$) is a vector-space over $\mathbb{R}$. As an exercise, it should be shown that it is an infinite dimensional vector-space¹. Since it is a real vector-space, it is isomorphic to $\mathbb{R}^N$ for appropriate cardinality N , $N \geq |\mathbb{N}|$. However, this space is *certainly* not $\mathbb{R}^{\mathbb{N}}$, since $\mathbb{R}^n$ compromises all real-sequences that are unbounded. So we would imagine some much larger cardinality, perhaps $\mathbb{R}^{\mathbb{R}}$. However, we run into many problems:

1. $\mathbb{R}^{\mathbb{R}}$ would be indistinguishable to L^1 , L^2 , L^∞ , and so on by what we've done.
2. If you think of a basis as a “super-dense” set, where instead of sequences converging to any point via the dense set, we only require a *finite* sequence (i.e. a function $f : X \rightarrow \mathbb{R}^{\mathbb{R}}$ where $|X| < \infty$), and not only that but also require the converge be to be done *linearly*. Even more powerfully, it requires that each element in the dense set be [linearly] *independent* of each other. Over-all, this is a huge huge ask. How would you even find a basis so that every element of $L^2(\mathbb{R})$ is represented by a finite linear combination of linearly independent basis elements??

Overall, it seems that the free-module intuition doesn't work anymore². Instead, when we think that a space is a vector-space, we should think of it in terms of the binary operation being defined and the scalar multiplication being defined. For example, if V is some infinite dimensional vector-space, that means that we know we can piece together elements inside it quite easily to get another element via $+$, and any element can be scaled by some constant. This ability to manipulate functions in this way should be what we should think of when thinking about infinite dimensional vector-spaces.

So how then do we intuit infinite dimensional vector-spaces if not the universal property of free module? Is there another universal property? Well, it turns out that in infinite dimensions, the situation is much more complicated. We will essentially find out how we deal with different types of infinite-dimensional vector-spaces by adding extra structure (in particular, a norm or an inner product), that will give us size or “shape” information for the vectors, which introduces a new notion of “finiteness” into our analysis (or even better a new way of comparing vectors to work with). Notice that since we are introducing a norm/inner-product, we are working with topologies. Unlike the finite dimensional case, where given a fixed dimension all norms are the same³, in infinite dimensions we will have vector-space with *different norms* or *different inner products* (and hence, that give us *different* topologies, unlike in the finite-dimensional case where there is only one “reasonable” topology which we may define that keeps our intuition of size or angle the same). This has many consequences we

¹answer: the functions $\chi_{[n,n+1)}$ are linearly independent for all $n \in \mathbb{N}$

²We will show how we can get something as close as possible to this intuition when we study Hilbert Spaces in chapter ref:HERE

³up to scaling, however this does not matter as a scaled norm produced the same topology

shall explore, for example in the finite dimensional case all linear maps are continuous, but not so in the infinite dimensional case (there will always be non-continuous linear maps). Thus, we may properly think of infinite dimensional vector-spaces as different than what we study in linear algebra, which doesn't offer the right tools to study such spaces. In fact, the place that does give us the right tools to figure out how to estimate or approximate things to limit the problem of infinities is *analysis*. Hence, we will essentially always require that all our spaces are *complete* in order to work with limits without any issues.

1

Preliminary Chapter

It shall be assumed that point-set topology, Measure Theory, L^p spaces, and optionally Distributions are known (at least up to the level presented in Folland). This chapter will cover:

1. Axiom of Choice
2. Nets and Filters
3. Important Compactness Results (ex. Stone-Weiestrass,)

1.1 Axiom of Choice

Look into this explanation of truth in mathematics, looks good!:

In mathematics, we are constantly making choices. We can make algorithmic choices (for example, if you're given the number n , choose the number $n + 1$), or we make arbitrary choices (choose some natural number n). The fact that we can make an arbitrary choice is uncontroversial in the axioms mathematicians used, that is ZF (short for Zermelo-Frankel axioms). So too is making a finite number of choices. These two "abilities" were also uncontroversial in other foundational formulations of mathematics, not just ZF. It might seem very natural then to be able to make an infinite, or more generally an arbitrary (including uncountably infinite or other cardinalities) amount of choices. However, making an arbitrary amount of choices turns out to have some very strange consequences. The German mathematician Ernst Zermelo axiomatically used the fact that you can make an arbitrary number of choices (which we will soon define what this means mathematically) to prove a very controversial conjecture called the *well-ordering Theorem* (which we shall cover later in this appendix), sparking a debate on whether mathematics should allow for arbitrary choice. One famous "counter-example" developed to "prove" that we shouldn't allow arbitrary choice is a proof by Banach and Tarski that you can take a sphere, partition it into 6 subsets (which uses the axiom of

choice), rotate them in certain ways, and recombine them to form 2 identical spheres! Importantly, we did *not* add any more points, yet we have somehow “doubled” the surface area of the points we’ve chosen ¹! This is called the *Banach-Tarski Paradox*.

Though this result is strange, as time went on, more and more mathematicians found that making an arbitrary number of choices produces some “natural” results. For example, the fact that every vector-space has a basis comes down to the ability of making an arbitrary number of choices. Another example (HERE: from that stackexchange, I think it’s the one saying “if $|A| = |B|$, then we need the AC to say there is an injective function from A to B ”). In 1963, a mathematician named Paul Cohen proved that the making an arbitrary number of choices is independent of the standard system of axioms ZF, meaning if we assume the we can do an arbitrary number of choices, it will not cause any more contradiction and ZF would already cause ², and if we assume we can’t make an arbitrary number of choices, it will *also* not cause any more contradiction than ZF would already cause. Through this result, most mathematicians now accept the axiom of choice as part of the standard model we use, which is represented as ZFC (ZF + the axiom of Choice).

In Algebra, the axiom of choice most often appears in a special form called *Zorn’s Lemma*. The aim of this chapter is to be able show that Zorn’s lemma is equivalent to the axiom of choice.

1.1.1 Build-up

Definition 1.1.1: Cartesian Product and Choice Function

Let $\{A_i\}_{i \in I}$ be any family of sets. Then the *Cartesian product* of this family, written $\prod_{i \in I} A_i$, is the set of all function $x : I \rightarrow \bigcup_{i \in I} A_i$, where $x(i) \in A_i$ for all $i \in I$. The function x is called a *choice function*.

Though we’ve defined what the cartesian product is, it does not mean that the function x exist! Naturally, if $I = \emptyset$ or if $A_i = \emptyset$ for some $i \in I$ (since then $\prod_{i \in I} A_i = \emptyset$), then this function exists. If $I \neq \emptyset$ or $A_i \neq \emptyset$, then it turns out the existence of this function is independent of the ZF axioms! Thus, the axiom of choice was formulated:

Axiom 1.1.1: Axiom of Choice

The cartesian product of any non-empty family of non-empty sets is a non-empty set. In particular, if $\{A_i\}_{i \in I}$ is a family of sets such that $I \neq \emptyset$ and $A_i \neq \emptyset$ for each $i \in I$, then there exists at least one choice function for $\{A_i\}_{i \in I}$.

Since the cartesian product exists (axiomatically), we will make some shorthand notation for it similar to that defined in chapter ?? . If $\{A_i\}_{i \in I}$ is a family of sets, let $A^I = \prod_{i \in I} A_i$ where $A_i = A$ for each $i \in I$. If $|I|$ is finite, and so $|I| = n$ for some $n \in \mathbb{N}$, then we write A^n instead. If $|I|$ is countable, we sometimes use the A^∞ or $A^\mathbb{N}$ notation. the elements of A^n can be represented as n -tuples, that is

$$A^n = \{(a_1, a_2, \dots, a_n \mid a_i \in A)\}$$

¹For those who have taken measure theory, the chosen subsets turn out to be *unmeasurable* sets, and so the concept of surface area does not apply to these sets which allows us to produce this strange result

²The careful wording here is because it has *not* been proven that the axioms of ZF don’t contradict each other – this remains an open problem in mathematics

and the elements of $A^{\mathbb{N}}$ can be represents as sequences

$$A^{\mathbb{N}} = \{(a_1, a_2, \dots, a_n, \dots) \mid a_i \in A\}$$

Example 1.1: Cartesian Products

1. An example we've already encountered is $\mathbb{R}^n$ and $\mathbb{R}^{\mathbb{N}}$. This cartesian product is called *Euclidan Space*
2. Let $A = \{0, 1\}$. Then $A^{\mathbb{N}}$ is the set of all sequences with either 0 or 1 in each component. If we change notation a bit and let $A = \mathbb{Z}/2\mathbb{Z}$, and define addition point-wise, then this would be a group

(words here)

Definition 1.1.2: Partially Ordered Sets

1. Let P be a non-mepty set. A *partial order* on P is a relation $\leq$ ($\leq \subseteq P \times P$) on P such that

- (a) **(reflective)** $x \leq x$ for all $x \in A$
- (b) **(antisymmetric)** if $x \leq y$ and $y \leq x$, then $x = y$
- (c) **(transitive)** if $x \leq y$ and $y \leq z$ then $x \leq z$

Where we usually say P is a partiallly ordered set under the orering $\leq$ or that A is partially ordered by $\leq$.

If $\leq$ also satisfies

- (d) **(trichotomy)** If $x, y \in P$, then $x \leq y$ or $y \leq x$,

then $\leq$ is called a *total* (or linear, simple, complete) ordering on P . If further

- (e) **(Well-Ordering)** For all $\emptyset \neq A \subseteq P$, there exists an element $a \in A$ such that $a \leq x$ for all $x \in A$

then we say that $\leq$ is a *well-ordering* of P , or that P is well-ordered by $\leq$

2. Let $(P, \leq)$ be a partially ordered set and let $A \subseteq P$. An element $u \in P$ is called an *upper bound* for A if for all $x \in A$, $x \leq u$. Similarly, $l \in P$ is called a *lower bound* if for all $x \in A$, $l \leq x$.
3. Let $(P, \leq)$ be a partially ordered set and let $A \subseteq P$. An element $m \in P$ is called a *maximal element* of P if whenever $m \leq x$ with $x \in P$, then $m = x$. Similarly, an element $m \in P$ is called a *minimal element* if whenever $x \leq m$ for $x \in P$, then $x = m$.
4. Let $(P, \leq)$ be a partially ordered set. Then a subset $C \subseteq P$ is called a *chain* in P if C is totally ordered under the order relation of $\leq$, that is, if $x, y \in C$, then either $x \leq y$ or $y \leq x$.

Note that it's called a *partial* order because not all elements *necessarily* need to have a relation between them: if $x, y \in P$ and $\leq$ is a partial order on P , then it need not be the case that $x \leq y$ or $y \leq x$. If this were the case, we would call $\leq$ a *total order*.

In the following examples we'll show the difference between being maximal and being an upper bound. Importantly, it is possible that *not all* $x \in P$ have the relation $x \leq m$ (i.e. m is not necessarily the "largest" or "maximum" element)

Example 1.2: Partially Ordered Sets

1. Let S be an arbitrary set, and $P(S)$ be it's powerset. Then $(P(S), \subseteq)$ forms a partially ordered set. This set is *only* a partial order (it is not a total order) if $|S| > 1$: for two distinct elements $x, y \in S$ $\{x\} \not\subseteq \{y\}$ and $\{y\} \not\subseteq \{x\}$. A chain in $(P(S), \subseteq)$ would look like:

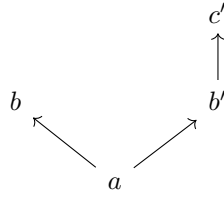
$$S_1 \subseteq S_2 \subseteq S_3 \subseteq S_4 \subseteq \dots$$

If $T \subseteq S$, then T has an upperbound U :

$$U = \bigcup_{x \in T} \{x\}$$

Furthermore, S is also the maximal element of $P(S)$.

2. Let $S = \{a, b, b', c'\}$ be the set, and $\leq$ be the relation visualized by the graph:



Then both b and c' are maximal elements. S has no upper bound, but it has a as a lower bound.

3. Let S be the collection of *proper subsets* of $\mathbb{N}$ ordered by $\subseteq$. Notice that the chain

$$\{1\} \subseteq \{1, 2\} \subseteq \{1, 2, 3\} \subseteq \dots$$

does not have an upper bound. However, S *does* have maximal elements. $(\mathbb{N} - \{n\}) \in S$ for any $n \in \mathbb{N}$ is a maximal element of S

4. Let $S = \mathbb{R}$ with the usual order $\leq$. Notice that $\leq$ is a *total order* in this case. some sets of $\mathbb{R}$ have upper bound (ex. $[0, 1]$ or $\{0, 1, 2, 3\}$), while other's don't ($\mathbb{Z}$, $(0, \infty)$, $(0, 1)$). $\mathbb{R}$ does not have maximal elements in this order.

When we say we give a partial order to an arbitrary set S , we will assume that this is the partial order on the power set given by $\subseteq$.

Next, we will build-up some facts to bootstrap the result

Definition 1.1.3: Finite Character

Let $\mathcal{F}$ be a family of sets. Then $\mathcal{F}$ is said to be a *family of finite character* if for any set S , $S \in \mathcal{F}$ if and only if each finite subset of S is in $\mathcal{F}$.

Lemma 1.1.1: Finite Character and Chains

Let $\mathcal{F}$ be a family of finite character and let $\mathcal{B}$ be a chain in $\mathcal{F}$. Then $\cup \mathcal{B} \in \mathcal{F}$

Proof :

Since $\mathcal{F}$ is of finite character, if we show that every finite subset of $\cup \mathcal{B}$ is an element of $\mathcal{F}$. Let $\{x_1, x_2, \dots, x_n\} = F \subseteq \cup \mathcal{B}$. Then there must exist $\mathcal{B}_1, \dots, \mathcal{B}_n \in \mathcal{B}$ such that $x_i \in \mathcal{B}_i$. Since $\mathcal{B}$ is a chain, there exists a $1 \leq j \leq n$ such that $\mathcal{B}_i \subseteq \mathcal{B}_j$, for all $i \in \{1, 2, \dots, n\}$. But then, $F \subseteq \mathcal{B}_j \in \mathcal{F}$. Since $\mathcal{F}$ is of finite character, and $\mathcal{B}_j \in \mathcal{F}$, $F \in \mathcal{F}$. Since F was arbitrary, then this holds true for any finite subset of $\cup \mathcal{B}$, and so $\cup \mathcal{B} \in \mathcal{F}$, as we sought to show.

1.1.2 Axiom of Choice Equivalences

Note that terms like “lemma” or “theorem” for the following names are due to historical precedents rather than actually representing a lemma or theorem, since all of the following will be equivalent to the axiom of choice.

Theorem 1.1.1: Turkey’s Lemma

Every nonempty family of finite character has a maximal member

Theorem 1.1.2: Hausdorff Maximality Principle

Every nonempty partially ordered set contains a maximal chain, i.e., every totally ordered subset is contained in a maximally totally ordered subset

Maximally totally ordered here means that if it is enlarged in any way, it does not remain totally ordered

Theorem 1.1.3: Zorn’s Lemma

Every nonempty partially ordered set in which each chain has an upper bound has a maximal element

Theorem 1.1.4: Well-Ordering Theorem

Every set can be well-ordered, i.e., if S is a set, then there exists some well-ordering $\leq$ on S

A well-ordering on a set S is a total order (i.e. a chain) on S with the additional property that every non-empty subset of S has a lower-bound. For example, given the usual orderings for the following examples, $\mathbb{N}$ is well-ordered (every subset has a lower-bound), while $\mathbb{Z}$ is not (namely, $\mathbb{Z}$ itself doesn’t

have a lower-bound). A trickier example would be $\mathbb{R}$. The set (a, b) for $a, b \in \mathbb{R}$, $a \neq b$, does *not* have a minimal element, and so $\mathbb{R}$ is not well-ordered with the usual ordering.

Theorem 1.1.5: Axiom of Choice Equivalences

The following are equivalent

1. The axiom of choice
2. Turkey's Lemma
3. The Hausdorff maximality principle
4. Zorn's Lemma
5. Well-ordering Theorem

Credit to the proof to

Proof :

Step (1) $\Rightarrow$ (2) will be the longest step requiring many technical details. the other steps will come much more naturally. The proof will also be long since we will not be using heavy machinery from a course in set-theory (for ex. we will not use transfinite induction [directly?? are we secretly using it?])

- (1) $\Rightarrow$ (2) For the sake of contradiction, let's say Turkey's lemma was false – there exists a non-empty family of sets of finite character that has no maximal element. let's say $\mathcal{F}$ was such a set. That means for every $A \in \mathcal{F}$, there exists a B such that $A \subsetneq B$. Let $\mathcal{O}_F = \{E \in \mathcal{F} \mid F \subsetneq E\}$. Then $\{\mathcal{O}_F \mid F \in \mathcal{F}\}$ is a set of sets indexed by $\mathcal{F}$. Thus, by the axiom of choice, there exists a choice function f defined on $\mathcal{F}$ such that $f(F) \in \mathcal{O}_F$ (i.e. $F \subsetneq f(F)$ for all F).

Using f , we'll define a new concept: a subfamily $\mathcal{F} \subseteq \mathcal{F}$ is f -inductive if:

- (a) $\emptyset \in \mathcal{F}$
- (b) if $A \in \mathcal{F}$ then $f(A) \in \mathcal{F}$
- (c) if $\mathcal{C} \subseteq \mathcal{F}$ is a chain, then $\cup \mathcal{C} \in \mathcal{F}$

Notice that $\mathcal{F}$ is itself an f -inductive set since: $\mathcal{F}$ is non-empty (by assumption), so some $A \in \mathcal{F}$ exists, and $\emptyset \subseteq A$ is a finite set, and so $\emptyset \in \mathcal{F}$ since $\mathcal{F}$ has finite character, fulfilling the first condition. The second condition holds true since the image of f is in $\mathcal{F}$, and the third condition holds by lemma 1.1.1. Notice too that if the family $\mathcal{F}_i \subseteq \mathcal{F}$ is f -inductive (for some indexing set I), then so too is $\bigcap_i \mathcal{F}_i$. Thus, define the smallest f -inductive set of $\mathcal{F}$:

$$\mathcal{F}_0 = \{A \in \mathcal{F} \mid A \in \mathcal{F} \text{ for every } f\text{-inductive subset of } \mathcal{F}\}$$

The fact that $\mathcal{F}_0$ is the smallest f -inductive set will allow us to force certain properties on to it. Our goal is to show that $\mathcal{F}_0$ *must* be a chain, which we will use to arrive at a contradiction.

To show that $\mathcal{F}_0$ is a chain, we will have to go through some technical steps. First, let $\mathcal{H} = \{A \in \mathcal{F}_0 \mid \text{if } B \in \mathcal{F}_0 \text{ and } B \subsetneq A \text{ then } f(B) \subseteq A\}$. The set $\mathcal{H}$ is the set

of all elements $A \in \mathcal{F}_0$ such that B element is an element of $\mathcal{F}_0$ and is properly contained in A , f of that elements is contained in or equal to A . However, some oddities occurs with this set too: note that $\mathcal{H}$ is non-empty since vacuously, $\emptyset \in \mathcal{H}$ (since $(\text{False} \Rightarrow \text{True})$ and $(\text{False} \Rightarrow \text{False})$ are both *true* statements!!) This oddity is necessary, since we are guaranteed that $\mathcal{H}$ is non-empty. Now, we'll show that if $A \in \mathcal{H}$ and $C \in \mathcal{F}_0$, then one of two relations *must* hold between A and C : either $C \subseteq A$ or $f(A) \subseteq C$. Notice we are forcing a subset relation between all elements of $\mathcal{F}_0$ and $\mathcal{H}$ – we'll use this to create a total order later. First, let's prove the last assertion:

Proof. Define $\mathcal{G}_A = \{C \in \mathcal{F}_0 \mid C \subseteq A \text{ or } f(A) \subseteq C\}$. By definition, $\mathcal{G}_A \subseteq \mathcal{F}_0$. If we show that $\mathcal{G}_A$ is f -inductive, then $\mathcal{F}_0 \subseteq \mathcal{G}_A$, and since $\mathcal{F}_0$ is the smallest f -inductive subset, it must be that $\mathcal{G}_A = \mathcal{F}_0$ which will tell us that any element $C \in \mathcal{F}_0$ must be either contained in A or contain $f(A)$. For notational convinience, let's label the condition $C \subseteq A$ as [1] and condition $f(A) \subseteq C$ as [2]

- (a) Since $\emptyset \in \mathcal{F}_0$, $\emptyset \subseteq A$, and so by [1] $\emptyset \in \mathcal{G}_A$
- (b) Let $C \in \mathcal{G}_A$. We must show that $f(C) \in \mathcal{G}_A$. Since $C \in \mathcal{G}_A$, then we have 3 possibilities (we're splitting [1] into 2 separate cases:
 - i. $C \subsetneq A$: Since $C \subsetneq A$ and $A \in \mathcal{H}$, then $f(C) \subseteq A$, so by [1] $f(C) \in \mathcal{G}_A$
 - ii. $C = A$: Since $C = A$ then $f(C) = f(A)$. Which is equivalent to $f(C) \subseteq f(A)$ and $f(A) \subseteq f(C)$. For our purposes, we care that $f(A) \subseteq f(C)$. By condition [2], $f(C) \in \mathcal{G}_A$
 - iii. $f(A) \subseteq C$, then since $C \subsetneq f(C)$ by definition of f , $f(A) \subseteq f(C)$, and so by condition [2], $f(C) \in \mathcal{G}_A$
 in all cases, $f(C) \in \mathcal{G}_A$
- (c) Let $\mathcal{C} \subseteq \mathcal{G}_A$ be a chain in $\mathcal{G}_A$. We must show that $\cup \mathcal{C} \in \mathcal{G}_A$. Then, either for all $C \in \mathcal{C}$, $C \subseteq A$, in which case $\cup \mathcal{C} \subseteq A$, and so by [1] $\cup \mathcal{C} \in \mathcal{G}_A$, or there is some $C \in \mathcal{C}$ $f(A) \subseteq C \subseteq \cup \mathcal{C}$, in which case by [2] $\cup \mathcal{C} \in \mathcal{G}_A$. In both cases, the chain is in $\mathcal{G}_A$

Thus, $\mathcal{G}_A$ is indeed f -inductive for any A , and so $\mathcal{F}_0 = \mathcal{G}_A$

□

Using the previous result, we will force one more important result: $\mathcal{H} = \mathcal{F}_0$.

Proof. As before, it is already given that $\mathcal{H} \subseteq \mathcal{F}_0$. If we show $\mathcal{H}$ is f -inductive, we'll have that $\mathcal{H} = \mathcal{F}_0$:

- (a) As mentioned before, $\emptyset$ has no proper subset, so $(F \Rightarrow T)$ is vacuously true, meaning $\emptyset \in \mathcal{H}$
- (b) Let $A \in \mathcal{H}$. We must show $f(A) \in \mathcal{H}$, that is if $B \in \mathcal{F}_0$ and $B \subsetneq f(A)$ then $f(B) \subseteq f(A)$. Since $B \in \mathcal{F}_0 = \mathcal{G}_A$, then either $B \subseteq A$ or $f(A) \subseteq B$. Since $B \subsetneq f(A)$, it cannot be the latter, thus we must have that $B \subseteq A$. From here, we have two possibilities:
 - i. $B \subsetneq A$: If so, then since $A \in \mathcal{H}$, by definition $f(B) \subseteq A \subseteq f(A)$, where the last $\subseteq$ comes from the definition of f . Thus, $f(B) \subseteq f(A)$
 - ii. $B = A$: Then $f(A) = f(B)$, in particular $f(B) \subseteq f(A)$
 in either case, we get $f(B) \subseteq f(A)$

- (c) Let $\mathcal{C} \subseteq \mathcal{H}$ be a chain. To show $\cup \mathcal{C} \in \mathcal{H}$, it must be that for any $B \in \mathcal{F}_0$ where $B \subsetneq \cup \mathcal{C}$, then $f(B) \subseteq \cup \mathcal{C}$. We will do a sneaky trick for this proof using the law of excluded middle ^a: For each $C \in \mathcal{C}$, either “ $C \subseteq A$ for some A ”, or “ $A \subsetneq C$ for all A ” (which is it’s direct negation). If the latter was true then

$$B \subsetneq \cup \mathcal{C} \subseteq \cup \{f(A) \mid A \in \mathcal{C}\} \subseteq B$$

implying $B \subsetneq B$ – a contradiction. Thus, it must be that for some $A \in \mathcal{C}$, $B \subseteq A$. There are again two cases

- i. $B \subsetneq A$: Since $A \in \mathcal{H}$, then $f(B) \subseteq A \subseteq \cup \mathcal{C}$, so $f(B) \subseteq \cup \mathcal{C}$
- ii. $B = A$: Then we have $B \in \mathcal{H}$, so $\mathcal{G}_B$ is defined and $\mathcal{F}_0 = \mathcal{G}_B$. Since $\mathcal{G}_B$ is f -inductive, $\cup \mathcal{C} \in \mathcal{G}_B = \mathcal{F}_0$. Thus, either $\cup \mathcal{C} \subseteq B$ or $f(B) \subseteq \cup \mathcal{C}$, and since $\cup \mathcal{C} \subseteq B$ is not possible since $B \subsetneq \cup \mathcal{C}$, it must be that $f(B) \subseteq \cup \mathcal{C}$.

In either case, $f(B) \subseteq \cup \mathcal{C}$, showing that $\cup \mathcal{C} \in \mathcal{H}$, showing that $\mathcal{H}$ is f -inductive

Thus, $\mathcal{F}_0 = \mathcal{H}$ as we sought to show □

Finally, we get to combine all these facts. If $A \in \mathcal{F}_0 = \mathcal{H}$ and $B \in \mathcal{F}_0 = \mathcal{G}_A$, then either $B \subseteq A$ or $A \subseteq f(A) \subseteq B$. We couldn’t do this earlier since we didn’t know that $\mathcal{F}_0 = \mathcal{H}$ and $\mathcal{F}_0 = \mathcal{G}_F$ for any $F \in \mathcal{F}_0$. Thus, we have that $A \subseteq B$ or $B \subseteq A$, showing that $\mathcal{F}_0$ is a chain. We can now define $M = \cup \mathcal{F}_0$. Since $\mathcal{F}_0$ is f -inductive, then by the 3rd condition $M \in \mathcal{F}_0$. By the 2nd condition:

$$\cup \mathcal{F}_0 = M \subsetneq f(M) \in \mathcal{F}_0$$

But $f(M) \subseteq \cup \mathcal{F}_0 = M$, so $M \subsetneq f(M) \subseteq M$ implying $M \subsetneq M$ – a contradiction. The source of the contradiction was assuming an f existed which let’s as choose an ever bigger value, and so no such function can exist, meaning for some A , there does not exist a B such that $A \subsetneq B$, but then A is maximal. And so, we have that (1) $\Rightarrow$ (2)

- (2) $\Rightarrow$ (3) Let $(P, \leq)$ be a non-empty partially ordered set. We want to show that $(P, \leq)$ contains a maximal chain (i.e. a chain C that cannot be enlarged further without loosing its total order). Let $\mathcal{F}$ be the set of all chains in P , ordered by inclusion (so if $P = A \leq B$ is a chain and $Q = A \leq B \leq C$, then $P \leq Q$). This set is non-empty, since $\emptyset \in \mathcal{F}$ and $\{x\} \in \mathcal{F}$ for all $x \in P$. Furthermore, Since any finite subchain of any infinite chain in $\mathcal{F}$ would be in $\mathcal{F}$, $\mathcal{F}$ has finite character. Thus, by Turkey’s Lemma, $\mathcal{F}$ has a maximal element, meaning P has a maximal chain.
- (3) $\Rightarrow$ (4) Let $(P, \leq)$ be a non-empty partially ordered set where each chain has an upper bound. Then by the Hausdroff Maximality Principle, there exists a maximal chain $M \subseteq P$. Since M is a chain, it has an upper-bound – let’s say m . Then m is must be a maximal element of P : If there exists some $x \in P$ such that $m \leq x$ and $m \neq x$, then $M \cup \{x\}$ is a chain that is properly included in M , but M is maximal – a contradiction. Thus, m is maximal in P
- (4) $\Rightarrow$ (5) Let S be any set, and let $\mathcal{L}$ be the family of all well-ordered set $(W, \leq)$ such that $W \subseteq S$. This set is non-empty, since for each $x \in S$, $\{\{x\}, \{x, x\}\} \in \mathcal{L}$. In the upcoming proof, we will give an ordering to $\mathcal{L}$ and use Zorns Lemma to show that the maximal element will be $(S, \leq)$ with appropriate $\leq$
- Define $(W_1, \leq_1) \preceq (W_2, \leq_2)$ if either

- (a) $W_1 = W_2$ and $\leq_1 = \leq_2$ or
- (b) there exists a $a \in W_2$ such that $W_1 = \{x \in W_2 \mid x \leq_2 a, x \neq a\}$, and $\leq_2$ agrees with $\leq_1$ on W_1 ($\leq_1 \subseteq \leq_2$ ^b). In this case, W_2 is called a continuation of $(W_1, \leq_1)$

It should be relatively clear that $(\mathcal{L}, \preceq)$ is a partial ordering on $\mathcal{L}$. Note that it is not a total ordering: If W_i and W_j do not have the same minimal element, then $W_i \not\preceq W_j$ and $W_j \not\preceq W_i$. This means that any chain of elements will end up with the same minimal element. Now, let $\ell = \{(W_i, \leq_i)\}_{i \in I}$ be any non-empty chain in $\mathcal{L}$ (relative to $\preceq$). We want to show it must have an upper-bound. Set $W = \bigcup_{i \in I} W_i$ and $\leq = \bigcup_{i \in I} \leq_i$ (see definition ??). Then $(W, \leq)$ is a total order (see exercise ref:HERE). We'll show it is a well-ordering. Let A be a non-empty subset of W . Since it's non-empty, there exists some $i \in I$ such that $A \cap W_i \neq \emptyset$. Since $(W_i, \leq_i)$ is well-ordered, there is an element $a \in A \cap W_i$ such that $a \leq_i x$ for all $x \in A \cap W_i$. Suppose now that there is an element $b \in A$ such that $b \leq a$. Then by definition of $\leq$, it must be that $b \in W_i$, so $b \leq_i a$, so $b = a$. Thus, A has a smallest element a in $(W, \leq)$, meaning $(W, \leq) \in \mathcal{L}$. Furthermore, it should be clear that $(W, \leq)$ is an upper-bound for ℓ

Thus, by Zorn's Lemma, $\mathcal{L}$ has a maximal element $(W_0, \leq_0)$. If $W_0 = S$, we're done — $\leq_0$ is our well-ordering of S . If $W_0 \neq S$, let $z \in S \cap W_0$. Define $(W_0 \cup \{z\}, \leq)$, where

$$\leq = \leq_0 \cup \{(x, z) \mid x \in W_0 \cup \{z\}\}$$

that is, place z after everything in W_0 . Then we can see that $(W_0 \cup \{z\}, \leq) \in \mathcal{L}$ and $(W_0, \leq_0) \preceq (W_0 \cup \{z\}, \leq)$ and $(W_0 \cup \{z\}, \leq) \not\preceq (W_0, \leq_0)$. However, this contradicts the maximality of $(W_0, \leq_0)$, and so it must be that $W_0 = S$, showing that S does indeed have a well-ordering.

- (5) $\Rightarrow$ (1) Let $\{A_i\}_{i \in I}$ be any non-empty family of non-empty sets, and let $S = \bigcup_{i \in I} A_i$. By the well-ordering principle, there exists an $\leq$ that well-orders S . Now, for each $i \in I$, let $f(i)$ be the smallest element of A_i (relative to $\leq$). Then f is a choice function on the family $\{A_i\}_{i \in I}$

^aFor every proposition P , either P is true or $\neg P$ is true. This is implied by ZF

^bIf these seems strange, review the technical definition of a relation in section ??

In this textbook, any non-constructive proof shall ultimately be relying on the axiom of choice to conclude its validity. As an example: (perhaps also basis for inf dim? Challenge the reader to try?) (The proof of Schröder-Bernstein Theorem requires Zorn's lemma, so I can show it here, it's in Folland's Real Analysis preliminary chapter)

(maybe also show Schröder-Bernstein's lemma in the finite case after introducing cardinality?)

1.2 Nets and Filters

1.3 Compactness

1.4 Topological Vector Spaces and Weak Topologies

If I move this here, I would have to re-haul the narrative to make it independent of the next chap

(This post seems like an excellent post to follow! [link here](#))

There are many topologies on which the operations are continuous (i.e. $+$ and scalar multiplication is continuous) that do not arise from norms. The topology on a vector-space induced by a norm is in fact quite strong. In general, such a space (in infinite dimensions) doesn't have many compact set, and so we lose many of the nice results of working on compact or pre-compact sets. This is naturally unfavourable in analysis, since many arguments rely on properties of sequences converging. Looking at functional analysis as the study of sequences, norms induce a particular definition of convergence: $\|x - x_n\|_V \rightarrow 0$. However, there are many other notions of convergence that are weaker that are quite useful, for example:

1. pointwise convergence
2. convergence in measure
3. smooth convergence
4. convergence on compact sets

None of these types of convergence are induced by a norm. We will instead be induced new types of topologies on our vector-spaces. The two types of topologies we will usually impose on V will be known as the *weak topology* and the *weak* topology*. We will define analogous topologies on $\text{Hom}_k(V, W)$ which somewhat confusingly will be named *strong operator topology* and *weak operator topology*.

Definition 1.4.1: Topological Vector Space (TVS)

Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$. Then V is a *topological vector space* (TVS) if the operation of addition and scalar multiplication are continuous. A topological vector-space is said to be *locally convex* if there is a base for the topology consisting of convex sets.

It should be verified that the translation map $x \mapsto x_0 + x$ and the dilation map $x \mapsto \lambda x$ are both homeomorphisms for any scalar λ and vector x_0 .

Example 1.3: Topological Vector Spaces

1. Let V be a normed vector-space with the topology induced by the norm. Then $\|c \cdot x\| \leq |c| \|x\|$ (in fact $=$) and $\|(x, y)\| \stackrel{!}{\leq} 2(\|x\| + \|y\|)$ where $\stackrel{!}{\leq}$ comes from the triangle inequality and the fact that $\|(x, y)\| = \max(\|x\|, \|y\|)$. Hence, V is a topological vector space. If $\|\cdot\|$ was a quasi-norm so that we replace the triangle inequality with $\|x + y\| \leq K(\|x\| + \|y\|)$ then a quasi-normed space V is still a topological vector space.

2. Let V be a semi-normed space so that $\|x\| = 0$ does not necessarily imply $x = 0$. Then V is a topological vector space with the topology induced by the semi-norm. V is Hausdorff if and only if the semi-norm is a norm.
3. Any subspace $W \subseteq V$ of a topological vector space V is a topological vector space with the subspace topology. Similarly for products and quotients.
4. Let V be a vector-space and $(\mathcal{T}_\alpha)_{\alpha \in A}$ be a collection of topologies on V that makes V a topological vector space. Let $\mathcal{T} = \langle \mathcal{T}_\alpha \rangle_{\alpha \in A}$ be the topology generated by $\bigcup_{\alpha \in A} \mathcal{T}_\alpha$ (this is the weakest topology that contains each $\mathcal{T}_\alpha$). Then $(V, \mathcal{T})$ is also a topological vector space. Importantly, a sequence (resp. net) $x_n \rightarrow x$ in $\mathcal{T}$ if and only if $x_n \rightarrow x$ in $\mathcal{T}_\alpha$ for all $\alpha \in A$ (recall that $x_n \rightarrow x$ in a general topology space if for any neighborhood U of x , there exists an N such that for all $n \geq N$, $x_n \in U$). Commonly, we will have $(\mathcal{T}_\alpha)_{i=1}^\infty$ be generated by countable collection of semi-norms. A complete space generated by a countable collection of semi-norms is called a *Fréchet space*.
5. Let $T : V \rightarrow W$ be a linear map between two vector spaces. Let V be a topology induced by a family of semi-norms $(\|\cdot\|_{V_\alpha})_{\alpha \in A}$ and W the topology induced by a family of semi-norm $(\|\cdot\|_{W_\beta})_{\beta \in B}$. Then T is continuous if and only if for each $\beta \in B$, there exists a finite subsets $A_\beta \subseteq A$ and a constant C_β such that

$$\|T(x)\|_{W_\beta} \leq C_\beta \sum_{\alpha \in A_\beta} \|x\|_{V_\alpha}$$

for all $x \in V$.

6. (Pointwise Convergence) Let X be a set and $\mathbb{C}^X$ the collection of all complex valued functions $f : X \rightarrow \mathbb{C}$. Then $\mathbb{C}^X$ is a complex-vector-space. For each $x \in X$, define the seminorm:

$$\|f\|_x := |f(x)|$$

Then the topology generated by all of these seminorms is the *topology of pointwise convergence* on $\mathbb{C}^X$ (it is also the product topology on this space). A sequence $(f_n) \subseteq \mathbb{C}^X$ converges to f in this topology if and only if it converges pointwise. Note that if X has more than one point, then none of the semi-norms individually generate a Hausdorff topology, but taken all together the induced topology is Hausdorff.

7. (Uniform Convergence) Let X be a topological space and $C(X)$ the set of complex-valued continuous function $f : X \rightarrow \mathbb{C}$. If X is not compact, then in general we would not expect functions in $C(X)$ to be bounded, so the sup-norm will not necessarily make $C(X)$ into a normed vector-space. Nonetheless, we can still define “balls” $B_r(f) \subseteq C(X)$ by:

$$B_r(f) := \left\{ g \in C(X) \mid \sup_{x \in X} |f(x) - g(x)| \leq r \right\}$$

these do form a topological structure on $C(X)$, but not a topological vector space structure since scalar multiplication is no longer continuous. Nonetheless, we have that $(f_n) \subseteq C(X)$ converges in this topology to a limit $f \in C(X)$ if and only if f_n converge uniformly to f . Thus $\sup_{x \in X} |f_n(x) - f(x)|$ is finite for sufficiently large n and converges to 0 as $n \rightarrow \infty$.

8. (Uniform Convergence on Compact sets) Let X and $C(X)$ be as in the above example. For every compact subset $K \subseteq X$, define a seminorm:

$$\|f\|_{C(K)} = \sup_{x \in K} |f(x)|$$

The topology generated by all these seminorms as K ranges over all compact subsets of X is called the *topology of uniform convergence on compact sets*. It is stronger than the topology of pointwise convergence but weaker than the topology of uniform convergence, and is frequently used in complex analysis. A sequence $(f_n) \subseteq C(X)$ converges to $f \in C(X)$ in this topology if and only if f_n converges uniformly to f on each compact set. Is the resulting space a topological vector space?

9. (Smooth Convergence) Let $C^\infty([0, 1])$ be the space of smooth functions $f : [0, 1] \rightarrow \mathbb{C}$. Then we can define the C^k -norm on this space for any non-negative integer k by the formula:

$$\|f\|_{C^k} := \sum_{i=1}^k \sup_{x \in [0, 1]} |f^{(i)}(x)|$$

where $f^{(i)}$ is the i th derivative of f . The topology generated by all C^k norms $k = 0, 1, 2, \dots$, is the *smooth topology*. A sequence f_n converges in this topology to a limit f if $f_n^{(i)}$ converges uniformly to $f^{(i)}$ for each $i \geq 0$. Is this a topological vector space?

10. (Convergence in measure) Let $(X, \mathcal{M}, \mu)$ be a measure space and $L(X)$ the set of measurable functions $f : X \rightarrow \mathbb{C}$. Then the sets

$$B_{r, \epsilon}(f) := \{g \in L(X) \mid \mu(\{x \mid |f(x) - g(x)| \geq r\}) < \epsilon\}$$

for $f \in L(X)$, $\epsilon, r > 0$ forms the basis for a topology that makes $L(X)$ into a topological vector space. A sequence $(f_n) \subseteq L(X)$ converges to f in this topology if and only if it converges in measure.

11. Let $[0, 1]$ be given the usual Lebesgue measure. Then show that $L^\infty([0, 1])$ cannot be given a topological vector space structure in which a sequence $(f_n) \subseteq L^\infty([0, 1])$ converges to f in this topology if and only if it converges a.e.
12. Let V be an $\mathbb{R}$ -vector space. a subset $U \subseteq V$ is called *algebraically open* if

$$\{t \in \mathbb{R} \mid x + tv \in U\}$$

is open for all $x, v \in V$. Then any open set in a topological vector space is algebraically open. Note that there are algebraically open sets that are not open (take any dense subset of S^1 , say D , and $\mathbb{R} \setminus \{D\}$.) This larger set of algebraically open sets forms a topology, but this topology does not give the $\mathbb{R}^2$ structure of a topological vector space.

13. Let X be a LCH space. On $\mathbb{C}^X$, the topology of uniform convergence on compact sets is defined by the seminorms $p_K(f) = \sup_{x \in K} |f(x)|$ as K ranges over compact subsets of X . If X is σ -compact and $\{U_n\}$ are as in proposition ref:HERE (folland p. 168), this topology is defined by the seminorms $p_n(f) = \sup_{x \in \overline{U_n}} |f(x)|$. In this case, $\mathbb{C}^X$ is easily seen to be complete, so it is a Fréchet space. By proposition ref:HERE, so is $C(X)$.

14. The space $L^1_{\text{loc}}(\mathbb{R}^n)$ is a Fréchet space with the topology defined by the seminorms $p_k(f) = \int_{|x| \leq k} |f(x)| dx$ (completeness follows from completeness of L^1).
15. Here is another useful place where TVS are defined. Let's say we want to study a space in which d/dx is continuous (and hence bounded) on $C^\infty([0, 1])$. However, this is essentially impossible to define the topology through a norm. Indeed, if $f_\lambda(x) = e^{\lambda x}$, then $(d/dx)f_\lambda = \lambda f_\lambda$, and so $\|d/dx\| \geq \lambda$ for all λ , no matter what norm is used on $C^\infty([0, 1])$. There are a few ways to rectify this:
- (a) One can consider differentiation as an unbounded operator from V to W where W is a suitable Banach space and V is a dense subspace of W (exercise 30 in Folland)
 - (b) One can consider differentiation as a bounded linear map from one Banach space V to a different one W such that $V = C^k([0, 1])$ and $W = C^{k-1}([0, 1])$ (exercise 9)
 - (c) One can consider differentiation as a continuous operator on a locally convex space V whose topology is not given by a norm. It is easy to construct families of seminorms on the space of smooth functions so that differentiation becomes continuous almost by definition. For example, the seminorms:

$$p_k(f) = \sup_{0 \leq x \leq 1} |f^{(k)}(x)|$$

where $k \in \{0, 1, 2, \dots\}$ makes $C^\infty([0, 1])$ into a Fréchet space (the completeness is proved in exercise 9), and d/dx is continuous by proposition 5.15, since $p_k(f') = p_{k+1}(f)$. Other examples are in exercise 45 in chapter 9 of Folland.

Proposition 1.4.1: Topological Vector Space And Hausdorff

Let V be a topological vector space. Then V is Hausdorff if and only if $\{0\}$ is closed

Proof :
exercise

We may ask what properties of Banach spaces extend to topological vector spaces. Some nice properties of Banach spaces include the uniform boundedness principle and the open mapping theorem. These can be extended to the slightly larger class of *Fréchet spaces* that we mentioned in the examples above. We will single them out and talk about them in the next section.

If V is a Topological vector space, we can take V^* to be the set of all continuous functionals from V to the base field $k \in \{\mathbb{R}, \mathbb{C}\}$. Since V has no norm, V^* has no norm topology. We can however still define interesting topologies on it. This topology is dual to a weaker topology we can put on V :

Definition 1.4.2: Weak And Weak* Topologies

Let V be a topological vector space and V^* its dual. Then

1. the *weak topology* on V is the topology generated by the seminorms $\|x\|_\lambda = |\lambda(x)|$ for all $\lambda \in V^*$
2. the *weak* topology* on V^* is the topology generated by the seminorms $\|\lambda\|_x := |\lambda(x)|$ for all $x \in V$

If V is also a normed vector-space, then the weak topology on V is weaker than the norm topology. If V is a normed space, then so is V^* and $(V^*)^*$. the weak* topology on V^* makes V^* into a topological vector space, and is a topology weaker than the weak topology on V^* (which is defined using the double dual $(V^*)^*$). When V is reflexive ($V \cong V^{**}$), then the weak and weak* topologies on V^* are in fact equivalent.

Definition 1.4.3: Weak Convergence

Let V a vector-space endowed with the weak topology. Then a sequence $(x_n) \subseteq V$ that converges in V with the weak topology is called *weakly convergent* and is said to *weakly converge* to a point x . A sequence weakly converges $x_n \rightarrow x$ if and only if

$$\lambda(x_n) \rightarrow \lambda(x) \quad \forall \lambda \in V^*$$

and is usually denoted as $x_n \rightharpoonup x$. Similarly, $\lambda_n \rightharpoonup \lambda$ if $\lambda_n(x) \rightarrow \lambda(x)$ for all $x \in V$ (λ_n , as a function of V , converges pointwise to λ)

Proposition 1.4.2: Normed Vector Spaces Weak Topology

Let V be a normed vector-space. Then the weak topology on V and the weak* topology on V^* are both Hausdorff. In particular, we conclude that in the weak and weak* limits, when they exist they are unique

Proof :

Use the Hahn-Banach Theorem

Example 1.4: Norm, Weak, Weak* Convergence

Let $V = c_0(\mathbb{N})$, $V^* = \ell^1(\mathbb{N})$, and $V^{**} = \ell^\infty(\mathbb{N})$. Let $e_1, e_2, \dots$ be the standard basis of any of these vector-spaces.

1. The sequence (e_i) converges weakly in V to zero, but not converge in norm in V
- 2.

On $B(X, Y)$, we can usually add some more topologies that allow for more sequences to converge:

Definition 1.4.4: Strong And Weak Operator Topology

Let $B(X, Y)$ be a Banach space. Then

1. The strong operator topology is the topology induced by the seminorms $T \mapsto \|T(x)\|_Y$ for all $x \in X$
2. the *weak operator topology* is the topology induced by the seminorm $T \mapsto |\lambda(T(x))|$ for all $x \in X$ and $\lambda \in Y^*$

Hence, a sequence $(T_n) \subseteq B(X, Y)$ converges in the strong operator topology to a limit T if and only if $T_n(x) \rightarrow T(x)$ strongly in Y for all $x \in X$, and T_n converges also in the weak operator topology. T_n converges to T in the operator norm topology if and only if $T_n(x)$ converges *uniformly* to $T(x)$ on bounded sets. Hence, we see that the weaker operator topology is weaker than the strong operator topology, which in turn is (somewhat confusingly) weaker than the operator norm topology.

We can rephrase the uniform boundedness principle for convergence now as follows:

Proposition 1.4.3: Uniform Boundedness Principle Reformulation

Let $(T_n) \subseteq B(X, Y)$ be a sequence of bounded linear operators from a Banach space X to a normed space Y , and let $T \in B(X, Y)$ be another bounded linear operator, and let D be a dense subspace of X . Then the following are equivalent:

1. T_n converges in the strong operator topology on $B(X, Y)$ to T
2. T_n is bounded in the operator norm (i.e. $\|T_n\|_{op}$ is bounded) and the restriction of T_n to D converges in the strong operator topology of $B(D, Y)$ to the restriction of T to D

One important result is that in the weak operator topology, unit balls are always compact:

Theorem 1.4.1: Alaoglu's Theorem

Let X be a normed space, and B^* the unit ball of X^* . Then B^* is compact in the w^* -topology (the weak operator topology)

Proof :

You can check analysis now, or for a cool measure theoretic way look at Folland.

2

Banach Spaces

One of the key problems a norm tries to fix is making sure there is still a reasonable “finiteness” to your space. Though a normed vector-space doesn’t have an ordering, there is something similar to it: in the same way the distance between any two real numbers is finite (i.e. finite) the *norm* of any element in a Banach space is *finite*. Thus, though a space might be infinite like L^1 , there is still some notion of finiteness allowing us to bring in many of the concepts we have worked with before and avoid the pathologies of the infinite.

2.1 Basic Definitions and Properties

Let k be either $\mathbb{R}$ or $\mathbb{C}$, let V or X denote a vector-space over k , with 0 representing the zero vector. Let kx denote the subspace of X spanned by x . If $M, N \subseteq X$, then $M + N = \{m + n \mid m \in M, n \in N\}$.

Definition 2.1.1: Semi-Norm And Norm

Let V be a vector-space. A *semi-norm* on V is a function $x \mapsto \|x\|$ from V to $[0, \infty)$ satisfying the properties:

1. $\|x + y\| \leq \|x\| + \|y\|$ (the triangle inequality)
2. $\|\lambda x\| = |\lambda| \|x\|$ (homogeneity of scalars)

If furthermore:

- (3) $\|x\| = 0$ if and only if $x = 0$

then the function is called a *norm*

Definition 2.1.2: Normed Vector-Space

Let V be a vector space. Then if there exists a function $\|\cdot\|$ that is defined on V , then $(V, \|\cdot\|)$ is called a *normed vector-space*.

If V is a normed vector-space, we can induce a metric $\rho(x, y) = \|x - y\|$, namely since:

$$\|x - z\| \leq \|x - y\| + \|y - z\| \quad \|x - y\| = \|(-1)(y - x)\| = \|y - x\|$$

Since any metric induces a topology, the induced topology by the metric induced by a norm $\|\cdot\|$ is called the *norm topology* on V . A nice consequence of the induced norm topology is that addition and scalar multiplication is continuous. From the axioms of a semi-norm, $\|0\| = 0$. There are semi-norms for which $x \neq 0$ but $\|x\| = 0$, as we'll soon show (recall that Hausdorff if and only if 0 is closed). Note that the norm $\|\cdot\| : V \rightarrow \mathbb{R}$ is evidently continuous in the topology it induces¹. If $x_n \rightarrow x$ in X , then for all $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that for all $n \geq N$, $\|x_n - x\| < \epsilon$. By the reverse triangle inequality:

$$|\|x_n\| - \|x\|| \leq \|x_n - x\| < \epsilon$$

showing that $\|x_n\|$ get's arbitrarily close, and so $\|x_n\| \rightarrow \|x\|$, i.e. $\|\cdot\|$ is continuous. This means that the pre-image of $\{0\}$ of $\|\cdot\|$ is closed. Furthermore, it is a vector-space. Denoting

$$K = \{x \in V \mid \|x\| = 0\}$$

Then if $x, y \in K$, $0 \leq \|x + y\| \leq \|x\| + \|y\| = 0$, and $\|cx\| = |c|\|x\| = |c|0 = 0$. Hence, we may consider V/K . What norm to put on V/K that is natural with respect to the projection map $\pi : V \rightarrow V/K$ will be addressed soon.

Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are said to be *equivalent* if there exists some $C_1, C_2 > 0$ st

$$C_1 \|\cdot\|_1 \leq \|\cdot\|_2 \leq C_2 \|\cdot\|_1$$

Equivalent norms define equivalent metrics and the same topology, and so the same Cauchy sequences. Recall that in finite dimensions, all norms are equivalent, meaning there is only one notion of distance. As we'll see, this is not the case in infinite dimensions.

Since we have a metric, we have a notion of completeness. As is the case when working with analysis, we will require our spaces to be complete:

Definition 2.1.3: Banach Space

Let V be a normed vector space. Then V is a *Banach space* if it is complete with respect to it's induced metric.

Example 2.1: Banach Spaces

1. Every finite dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ is a Banach space. In this space, all linear map are continuous.
2. Let K be a compact space, and take the set of all continuous function that vanish at infinity: $C_0(K)$. Then $C_0(K)$ is a Banach space with the operator norm (see definition 2.1.6). We

¹Note that in the metric space topology, continuous if and only if sequentially continuous

will later show (Banach–Mazur theorem) that all real separable Banach spaces are isometric (the isomorphisms in the category of Banach spaces) to a subspace of $C_0([0, 1])$, giving us a universal way of thinking about Banach spaces. If the space is not separable, if the smallest dense set D of a Banach space X has density of cardinality α , then X is isometric to $C_0([0, 1]^\alpha, \mathbb{R})^a$

3. If X is a topological space, then $B(X, k)$ and $BC(X, k)$ (all bounded functionals and bounded continuous functionals) form a Banach space with the uniform norm $\|f\|_u = \sup_{x \in X} |f(x)|$.
4. For p where $1 \leq p \leq \infty$, the space $L^p(X)$ for some measurable space X is a Banach space with the L^p -norm (recall that $L^p(\mu)$ is the set of equivalence classes functions that agree a.e.; if $L^p(\mu)$ represented only functions, then $\|\cdot\|_p$ would be a semi-norm)
5. Two other common examples of Banach spaces we will encounter later will be special spaces of differentiable functions, Banach Algebras, and Hilbert Spaces.

There are also examples of spaces that are not Banach Spaces:

1. Let X be a locally compact Hausdorff space and consider $C_c(X)$ with the ∞ -norm. Then it is not a Banach space, not being complete. It is easy to see that $C_c(X)$, the set of continuous bounded functions, is a Banach space and contains $C_c(X)$, and so it suffices to find the closure of $C_c(X)$ in $C_b(X)$. The closure turns out to be $C_0(X)$.
2. Let $U \subseteq \mathbb{R}^n$ be open and take $C_0(U, \mathbb{C})$. Then This space is *not* a Banach space
3. The space $H(U, \mathbb{C})$ of all holomorphic functions from U to $\mathbb{C}$ is *not* a Banach Space
4. The space $C_c^\infty(\mathbb{R}^n)$ is *not* a Banach space
5. (If you know some PDE's) The space of distributions $\mathcal{D}'$ is *not* a Banach Space

These are all examples of a weaker structure known as a *topological vector space* which we shall cover in section ??.

^aAn interesting result using Banach-Mazur's theorem by Luis Rodríguez-Piazza shows that the space of smooth functions with the uniform norm is isometric to the space of nowhere differentiable functions

There is a nice way to check whether a vector space V is a Banach space which requires the following definition (this essentially follows from the fact that continuity if and only if sequential continuity in a metric space and that $\|\cdot\|$ is continuous)

Definition 2.1.4: Absolutely Convergent

Let V be a normed vector space. and $\{x_k\}_k^\infty \subseteq V$ a sequence. Then the sequence is said to *converge absolutely* if

$$\sum_k^\infty \|x_k\| < \infty$$

Lemma 2.1.1: Cauchy and Absolute Convergence

Let V be a normed vector space. Then V is complete if and only if every absolutely convergent series in V converges.

The idea is that $\mathbb{R}$ has an order, making it easier to produce many arguments about sequences.

Proof :

Let's first say that X is complete and that $\sum_{n=1}^{\infty} \|x_n\| < \infty$. To show it converges, we take advantage of completeness and show there is a Cauchy sequence. In particular, let $S_n = \sum_{k=1}^n x_k$ be the partial sum. Then since the series is absolutely convergent, there for all $n > m$,

$$\|S_n - S_m\| \leq \sum_{k=m+1}^n \|x_k\| \rightarrow 0 \quad \text{as } m, n \rightarrow \infty$$

so the sequence $\{S_n\}$ is Cauchy, and so converges, so $\{x_k\}$ converges.

Conversely, let's say an absolutely convergent series converges, and choose any Cauchy sequence $\{x_k\}$. Since it's a Cauchy sequence, choose a subsequence such that

$$\|x_{k_{i+1}} - x_{k_i}\| < 2^{-i}$$

Take $y_1 = x_{k_1}$ and $y_i = x_{k_{i+1}} - x_{k_i}$ for $i > 1$ so that $\sum_{i=1}^n y_i = x_{k_n}$. Then notice that $\{y_i\}$ is in fact absolutely convergent:

$$\sum_{i=1}^{\infty} \|y_i\| \leq \|y_1\| + \sum_{i=1}^{\infty} 2^{-i} = \|y_1\| + 1 < \infty$$

So $\sum_{i=1}^{\infty} y_i = \lim x_{k_n}$ exists. Since the subsequence of every Cauchy sequence approaches the same "point" (just like convergent sequences), we see that $\{x_n\}$ also converges to the same point, showing that Cauchy sequences are in fact convergent sequences, as we sought to show.

Next, we would want to study which linear functions between two vector-spaces are *continuous*. It will *not* be the case that all linear functions will be continuous. The following builds up to the classification of continuous linear functions:

Definition 2.1.5: Bounded Linear Map

Let $T : X \rightarrow Y$ be a linear map between two normed vector space. Then if there exists some C such that for all $x \in X$

$$\|T(x)\| \leq C\|x\|$$

Then T is said to be *bounded*, or is a *bounded linear operator*.

Note how this is different then boundedness of a set (ex. the range is bounded if $\|T(x)\| \leq C$). This definition is would be useless to us since no nonzero linear map would satisfy it. All linear map over finite dimensional vector spaces clearly bounded (can you find the C ?), so this concept gets interesting in infinite dimesion

Proposition 2.1.1: Bounded $\Leftrightarrow$ Continuous

Let $T : X \rightarrow Y$ be a linear map between two normed vector spaces. Then the following are equivalent:

1. T is continuous
2. T is continuous at 0
3. T is bounded

Proof :

If T is the zero map, then we're done, so assume $T \neq 0$

(1) implies (2) by definition. For (2) implies (3), Since T is continuous at 0, for all $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|x - 0\| < \|x\| < \delta$, then $\|T(x) - T(0)\| = \|T(x) - 0\| = \|T(x)\| < \epsilon$. Choosing $\epsilon = 1$, we get that there exists a δ_1 such that

$$\|x\| < \delta_1 \rightarrow \|T(x)\| < 1$$

we now want to show there exists a C such that $\|T(y)\| \leq C\|y\|$ for all y in the domain. We will do this by taking advantage of linearity and shrinking element so that we can take advantage of continuity.

Define $x(y) = \frac{\delta_1}{2\|y\|}y$ so that

$$\|x(y)\| = \left\| \frac{\delta_1}{2\|y\|}y \right\| = \frac{\delta_1}{2} \frac{1}{\|y\|} \|y\| = \frac{\delta_1}{2} < \delta_1$$

Therefore, by continuity and linearity:

$$\|T(x(y))\| = \left\| T\left(\frac{\delta_1}{2\|y\|}y\right) \right\| = \frac{\delta_1}{2\|y\|} \|T(y)\| < 1$$

since this is true for all y , we if we re-arrange we have:

$$\|T(y)\| < \frac{2}{\delta_1} \|y\|$$

showing that T is indeed bounded.

Finally, if $\|T(x)\| \leq C\|x\|$, then for all $\epsilon > 0$, choose $\delta = C^{-1}\epsilon$ so that

$$\|T(y) - T(x)\| = \|T(y - x)\| \leq \epsilon$$

completing the proof.

Note that for $T \in B(X, Y)$ to be an isomorphism, we require that T^{-1} also be bounded². If $\|T(x)\| = \|x\|$ for all x , then T is called an *isometry*.

Since the topology is that of norms, then continuous linear functions between normed vector-spaces

²We will show using the open mapping theorem that it suffices to show T is continuous and bijective

are the homomorphisms, meaning they preserve the norm (to the degree of boundedness mentioned in the above proposition) and the vector-space structure (to the degree the map is injective). We will denote the set of bounded linear functions (which we will call linear operators) as $B(X, Y)$, $\mathcal{L}(X, Y)$, or $\text{Hom}_k(X, Y)$, where we understand that since X and Y are normed spaces $\text{Hom}_k(X, Y)$ refers to continuous linear maps.

Like with most fields of mathematics, we are more often more interested in the functions between spaces rather than the spaces itself, for it is function that ultimately give us information about the structure we are working with. Thus, if we can use the techniques we will be developing on $B(X, Y)$, we may gain many interesting results. For this to be the case, we require to introduce a norm on $B(X, Y)$. Naturally, we want that the norm on $B(X, Y)$ to be compatible with X and Y in some way, especially the codomain Y (as we usually investigate the structure of the image of a map). The following norm on $B(X, Y)$ will be shown to be exactly that:

Definition 2.1.6: Operator Norm

$B(X, Y)$ is a normed vector space with norm^a

$$\begin{aligned}\|T\| &= \sup_{\|x\|=1} \|T(x)\| \\ &= \inf \{C \mid \|T(x)\| \leq C\|x\| \text{ for all } x\} \\ &= \sup_{\|x\| \neq 0} \frac{\|T(x)\|}{\|x\|}\end{aligned}$$

If $\|T(x)\| = \|x\|$, then T is called an *isometry*

^athe third equality holds if $V \neq \{0\}$

In the special case where we have $B(X, k)$, we write V^* and call it the dual space with the *norm topology*. It is useful to establish when is $B(X, Y)$ a Banach space. Note that the operator norm is well-defined since each $f \in B(X, Y)$ is bounded.

Proposition 2.1.2: Completeness Of $B(X, Y)$

Let X, Y be normed vector-space and $B(X, Y)$ the normed vector-space of all bounded linear functions from X to Y . If Y is complete, so is $B(X, Y)$.

Proof :

Let (T_n) be a Cauchy sequence in $B(X, Y)$. For each $x \in X$, we have a Cauchy sequence $(T_n(x)) \subseteq Y$, and thus a limit vector which we will denote $T(x)$. This defines a function $T : X \rightarrow Y$ which is clearly linear. The inequality

$$\begin{aligned}\|T(x) - T_n(x)\| &= \lim_m \|T_m(x) - T_n(x)\| \\ &\leq (\limsup_m \|T_m - T_n\|)\|x\|\end{aligned}$$

shows that T is bounded and that $T_n \rightarrow T$.

Hence, V^* is always a Banach space, since $\mathbb{R}$ and $\mathbb{C}$ are complete. Next, we see what happens with common vector-space constructions like products and quotients.

Subspaces, Product spaces, Quotient Spaces

There are 3 equivalent norms we can put on the product of two normed vector-spaces:

1. $\|(x, y)\| = \max(\|x\|, \|y\|)$
2. $\|(x, y)\| = \|x\| + \|y\|$
3. $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$

The key to these being equivalent is that we are only multiplying by a finite number of spaces, a different strategy would have to be implemented for norms on infinite dimensional vector spaces (recall the different ℓ^p -norms put on the set of all sequences). For subspace, it is clear that if $W \subseteq V$ is a subspace of a normed vector space V , then W has an induced norm with respect to V . What is more interesting is if V is a Banach space and we want W to be a Banach space. We then require all sequences to converge. Since V is Hausdorff (even metrizable since there is a norm on V), we see that W is complete if W is a *closed subspace* with respect to the topology of V . For quotients V/W , we would like to define a new norm on V/W so that the natural projection map $\pi : V \rightarrow V/W$ is continuous. This leads us to the following definition:

Proposition 2.1.3: Banach Quotient Space

Let X be a normed vector space and $Y \subseteq X$ a subspace of X . Then $\pi : X \rightarrow X/Y$ is the quotient map. This induces the topology on X/Y by defining:

$$\|\pi(x)\| = \|x + Y\| = \inf_{y \in Y} \|x - y\|$$

which is a *seminorm* on X/Y . If Y is closed, then this is in fact a norm, and if furthermore X is Banach, then X/Y is a Banach space.

The intuition in my mind for this definition of quotient norm is to “eliminate” any effect of the quotienting space Y on any element of $x + Y$. Take for example $\mathbb{R}^2/\mathbb{R}$. Then the norm of $\|(x, y) + \mathbb{R}\|$ should really be $|x|$. Furthermore, π is naturally a continuous map since it is norm-decreasing, meaning $\|\pi(x)\| \leq \|x\|$ (simply choose $C = 1$).

Proof :

For subadditivity, we take advantage of the definition of infimum: for every $x_1, x_2 \in X$, there exists an $\epsilon > 0$ and elements $y_1, y_2 \in Y$ such that

$$\begin{aligned} \|\pi(x_1)\| + \|\pi(x_2)\| + \epsilon &\geq \|x_1 - y_1\| + \|x_2 - y_2\| \\ &\geq \|x_1 + x_2 - (y_1 + y_2)\| \\ &\geq \|\pi(x_1 + x_2)\| \end{aligned}$$

Since ϵ was arbitrary, we have subadditivity. Homogeneity is immediate since π is linear, and hence we have a seminorm.

If $\|\pi(x)\| = \|x + Y\| = 0$, there is a sequence $(y_n) \subseteq Y$ such that $\|x - y_n\| \rightarrow 0$. If Y is closed,

$\lim_n(y_n) = x \in Y$, meaning $\pi(x) = 0$. It follows that X/Y is a normed space if and only if Y is closed in X .

Now, say $(z_n) \subseteq X/Y$ is a Cauchy sequence. Then we can find a subsequence (z'_n) were

$$\|z'_{n+1} - z'_n\| < 2^{-n}$$

By induction we can choose $x_n \in X$ such that $\pi(x_n) = z'_n$ and $\|x_{n+1} - x_n\| < 2^{-n}$. If X is Banach, (x_n) converges to an element $x \in X$, and since π is “norm decreasing” (the norm the same or smaller), then (z'_n) converges to $\pi(x)$. Hence, (z_n) converges to $\pi(x)$, and so X/Y is complete, as we sought to show.

Proposition 2.1.4: Universal Property of Quotient Maps

If $T \in B(X, Y)$ where X and Y are normed spaces, and if $W \subseteq X$ is a closed subspace of X containing $\ker(T)$, then there exists an operator $\tilde{T} \in B(X/W, Y)$ defined pointwise as $\tilde{T}(\pi(x)) := T(x)$ such that

$$\|\tilde{T}\| = \|T\|$$

in particular, the following diagram commutes in **Ban**:

$$\begin{array}{ccc} X & \xrightarrow{T} & Y \\ \pi \downarrow & \nearrow \tilde{T} & \\ X/W & & \end{array}$$

Proof :

First, by elementary linear algebra we have that $\tilde{T}$ is an operator on X/W , which is a normed space by the earlier proposition. Since

$$\|\tilde{T}(\pi(x))\| = \|T(x)\| = \|T(x - z)\| \leq \|T\|\|x - z\|$$

for every $z \in W$, it follows from the definition of the quotient norm that

$$\|\tilde{T}(\pi(x))\| \leq \|T\|\|\pi(x)\|$$

hence, $\|\tilde{T}\| \leq \|T\|$. The reverse inequality $\|\tilde{T}\| \geq \|T\|$ is clear since Q is norm decreasing.

Banach spaces are closed under extensions:

Proposition 2.1.5: Extending Banach Spaces

Let X be a normed space and Y a closed subspace. Then if Y and X/Y are Banach spaces, then X is a Banach space.

Proof :

Let $(x_n) \subseteq X$ be a cauchy sequence. Since $\pi : X \rightarrow X/Y$ is norm-decreasing, $(\pi(x_n)) = (y_n) \subseteq X/Y$ is a cauchy sequence. Since X/Y is complete, it converges to a point $\bar{x} \in X/Y$. Next, by the definition of the quotient norm, for every x_n , there exists an n st

$$\|(x - x_n) - y_n\| \leq \frac{1}{n} + \|\pi(x - x_n)\|$$

Then we get:

$$\begin{aligned} \|y_n - y_m\| &= \|(y_n + x - x_n) - (y_m + x - x_m) + x_n - x_m\| \\ &\geq \frac{1}{n} \|\pi(x - x_n)\| + \frac{1}{m} \|\pi(x - x_m)\| + \|x_n - x_m\| \end{aligned}$$

which is clearly cauchy, and hence since Y is banach it is complete and so this sequence converges to y . Finally, it is easy to see that

$$\|x_n - (x + y)\| \leq \|x_n - (x - y_n)\| + \|y - y_n\|$$

showing that $x - n \rightarrow x + y$, as we sought to show.

Finite space naturally have none of the problems introduced in infinite dimensions and preserve all of their nice properties even when subspace of infinite dimensinoal vectorspace:

Proposition 2.1.6: Finite Dimensional Normed Space

Every finite-dimensional subspace Y of a normed space X is a Banach space, and consequently is closed in X . Moreover, if $\dim(Y) = n$, every linear isomorphism of $\mathbb{F}^n$ onto Y is a homeomorphism

Proof :

p. 46 Analysis now.

Working with infinite dimensional space dense subsets become of key interest. As a simple example, the smooth bump functions form a dense subset of L^p for all $1 \leq p < \infty$. The following simplifies for us how we may define functions from spaces given a known dense set:

Proposition 2.1.7: Defining Operator using Dense Sets

Let X and Y be banach spaces and X_0 is a dense subspace of X . Then every operator $T_0 \in B(X_0, Y)$ has a unique extension to an operator $T \in B(X, Y)$ and $\|T\| = \|T_0\|$

Proof :

For each $x \in X$, there is a sequence $(x_n) \subseteq X_0$ that converges to x : $x_n \rightarrow x$. Since T_0 is a bounded operator ($\|T_0(x_n)\| \leq C\|x_n\|$), it is clear that $(T_0(x_n))$ is a convergent sequence in Y ; let's say it converges to $T(x)$. It should be clear that if we choose another sequence converging to x and do this procedure, then we will get the same value of $T(x)$. Thus, the map $x \rightarrow T(x)$ is a operator

that is linear, extends T_0 , and by construction

$$\|T\| = \|T_0\|$$

an excellent example would be polynomial over a compact set are dense over continuous functions over compact sets, which themselves are dense over all L^p spaces. Hence in the compact case, for real or complex functions, it suffices to define our function over the polynomials. For the case of locally compact spaces, Note that smooth bump functions are dense in L^p , and since they have compact support a polynomial sequence uniformly converges to them. Hence, we may define a lot of operators on different real/complex functions spaces over locally compact spaces using polynomials.

For future reference, it will be important to remember that all normed spaces can be completed and that the norm between a normed space X and its completion $\overline{X}$ is isometric:

Proposition 2.1.8: Unique Extension To Banach Space

For each normed space X , there is a Banach space $\overline{X}$, uniquely determined up to isometric isomorphism, such that $\overline{X}$ contains X as a dense subspace

Proof :

p. 47 Analysis now

An important concept in any infinite dimensional space is to figure out which sets are compact. For example, you may recall in real analysis that we proved the Stone-Weierstrass theorem, or even simpler that in $\mathbb{R}^n$ a set is compact if it is closed and bounded. The following shows that subsets that are “almost finite-dimensional” are compact.

Proposition 2.1.9: Compact Sets In Banach Spaces

Let $K \subseteq V$ be a subset of a Banach Space V . Then the following are equivalent:

1. K is compact
2. K is sequentially compact
3. K is closed and bounded, and for every $\epsilon > 0$, K lies in the ϵ -neighborhood

$$\{x \in V \mid \|x - y\| < \epsilon \text{ for some } y \in W\}$$

of a finite dimensional subspace W of V .

Suppose further that there is a nested sequence $V_1 \subseteq V_2 \subseteq \dots$ of finite dimensional subspaces of V such that $\bigcup_{n=1}^{\infty} V_n$ is dense. Then the following statement is equivalent to the first three:

4. K is closed and bounded, and for every $\epsilon > 0$, there exists an n such that K lies in the ϵ -neighborhood of V_n

Proof :

exercise

Example 2.2: Compact Sets In Banach Spaces

Let $1 \leq p < \infty$ and take a set $K \subseteq \ell^p(\mathbb{N})$. In order for it to be compact in the norm topology, it needs to be closed, bounded, and also uniformly p^{th} -power integrable at spatial infinity, that is for every $\epsilon > 0$, there exists an $n > 0$ such that

$$\left(\sum_{m>n} |f(m)|^p \right)^{1/p} \leq \epsilon$$

for all $f \in K$. This shows that the moving bump set is not uniformly p^{th} power integrable and thus not a compact subset, even though it's closed and bounded.

This condition doesn't translate to the continuous equivalent $L^p(\mathbb{R})$. We also need some uniform integrability at a "finite scale", which would be described using harmonic analysis tools like Fourier transforms, and hence we simply mention it here without developing it further.

Given the Banach space $\text{End}_k(V)$, it is natural to have a second binary operation of composition and ask if it is also continuous (hence bounded). More generally, if $T \in B(X, Y)$, and $S \in L(Y, Z)$, then

$$\|ST(x)\| \leq \|S\|\|T(x)\| \leq \|S\|\|T\|\|x\|$$

and so $ST \in B(X, Z)$. In particular, that makes $B(X, X) = \text{End}_k(X)$ into an algebra, and when X is complete, $B(X, X)$ is a *Banach Algebra*. More generally:

Definition 2.1.7: Banach Algebra

Let X be a normed k -algebra where addition, multiplication, and scalar multiplication are continuous with respect to the induced topology. Then if X is complete, X is a *Banach Algebra*.

For the multiplication to be continuous, it suffices to check that $\|xy\| \leq \|x\|\|y\|$ (can you see why?). If X has a multiplicative unit, it is called *unital*. Not all Banach algebras have a multiplicative unit, $C_0(X)$ where X is a locally compact (hausdorff) space and $C_0(X)$ is the set of functions from X to $\mathbb{R}$ (or $\mathbb{C}$) that vanish at infinity is *not* unital. On the other hand, $C_b(X)$, the set of bounded functions with pointwise operations with the supremum norm *is* unital. We will work more with Banach Algebras later.

Exercise 2.1.1

1. Let $A, B \subseteq V$ be two closed subspaces. Show that $A + B$ need not be a closed subspace! However, if one of A or B is finite dimensional, then $A + B$ is closed.
2. Show that the open unit ball in X maps onto (by π) the open unit ball of X/Y , but a similar statement about closed unit balls holds only in special cases.
3. For the categorically inclined, it should be verified that the topology on X/Y is the quotient topology corresponding to the equivalent relation $\sim$ where $x_1 \sim x_2$ if and only if $x_1 - x_2 \in Y$.

2.2 Baire Category Theorem

In this section, we introduce some of the basic tools that allow us to work with functions between Banach spaces. Of keen interest to us is the *open mapping theorem*, the *closed graph theorem*, and the *uniform boundedness theorem*. The open mapping theorem makes it sufficient that a surjective continuous linear operator is open, which in particular means that if it is also bijective it is an isomorphism. The closed graph theorem allows us to show that a linear operator $T : X \rightarrow Y$ is continuous if the graph $\Gamma(T)$ is closed, i.e. if all sequences in the graph converge. This will lead to great simplifications in showing a function is continuous. Finally, the uniform boundedness theorem will give a powerful criterion a family of operators is uniformly bounded when their images or each separately (i.e. “point-wise”) bounded.

The underpinning results we need are about dense sets:

Proposition 2.2.1: Baire Category Theorem I

Let (A_n) be a sequence of open dense subset of a complete metric space (X, d) . Then $\bigcap_n A_n$ is dense in X .

Proof :

Let B_0 be any closed ball in X with radius $r > 0$. Then since A_1 is dense in X and is open, the set $A_1 \cap \text{int}(B_0)$ contains a closed ball B_1 with radius $2^{-1}r$. Since A_2 is an open set and B_1 is a closed ball, we can repeat this with $A_2 \cap \text{int}(B_1)$ and find a closed ball B_2 with radius $2^{-2}r$. By induction, we get a sequence of closed balls (B_n) in X such that $B_n \subseteq A_n \cap \text{int}(B_{n-1})$, each with radius $< 2^{-n}r$ for every n . Since X is complete (and Hausdorff, since metric topological are normal), there exists a unique point $\{x\}$ such that

$$\{x\} = \bigcap_n B_n \subseteq B_0 \cap \left(\bigcap_n A_n \right)$$

This shows that $\bigcap_n A_n$ intersects every nontrivial ball X , which implies density, completing the proof.

Corollary 2.2.1: Countable union of Empty Interior Sets

The countable union of closed sets with empty interior has empty interior

Proof :

exercise

Proposition 2.2.2: Baire Category Theorem II

Let $T : V \rightarrow W$ be a bounded linear operator between two Banach spaces, $T \in B(V, W)$. Then if the image of the unit ball $B_1(0)$ is dense in some ball $B_r(0)$ with $r > 0$ ($T(B_1(0))$ dense in $B_r(0)$), then

$$B_{(1-\epsilon)r}(0) \subseteq T(B_1(0))$$

for every $\epsilon > 0$

In a sense, If $T(B_1(0))$ is dense in some ball $B_r(0)$, then it is the “supremum” of $B_r(0)$

Proof :

Let $U = T(B_1(0))$. Since $T(B_1(0))$ is dense in $B_r(0)$, for any y and $\epsilon > 0$, there is a $y_1 \in U$ such that $\|y - y_1\| < \epsilon r$. Consider now ϵU . This ball is dense in $B_{\epsilon r}(0)$. Hence, there exists a y_2 such that $\|(y - y_1) - y_2\| < \epsilon^2 r$. Continuing by induction, we get a sequence (y_n) where $y_n \in \epsilon^{n-1}U$ and

$$\left\| y - \sum_{k=1}^n y_k \right\| < \epsilon^n r$$

Choose now $x_n \in V$ such that $T(x_n) = y_n$ and $\|x_n\| \leq \epsilon^{n-1}$. Then $\sum x_n = x \in X$ converges, and by definition $T(x) = y$. Since $\|x\| \leq \sum \epsilon^{n-1} = (1 - \epsilon)^{-1}$ ^a we have

$$(1 - \epsilon)^{-1}U \supseteq B_r(0)$$

completing the proof.

^arecall that $1 - b^n = (1 - b)(1 + b + b^2 + \dots + b^{n-1})$

Let us now use these results to prove a couple of very useful results. Before proceeding, the following are some very useful lemmas the reader should prove to gain an intuition on open/closed maps:

1. f is a closed map if and only if if a net $y_n \rightarrow y$, then $f^{-1}(y_n) \rightarrow f^{-1}(y)$.
2. f is open if and only if whenever $x \in X$ and $y_n \subseteq Y$ is a net that clusters at $f(x) \in Y$, then $f^{-1}(y_n)$ clusters in X (note that we don't necessarily get convergence due to the lack of closedness, which would include limit points). Note that this also means that if y_n clusters around $y \in Y$, then $f^{-1}(y_n)$ clusters around every point in $x \in f^{-1}(y)$. If f is continuous, then if $x_n \subseteq X$ clusters at a point x , then $f(x_n)$ clusters at $f(x) \in Y$
3. We can add a bit more to the above: f is open if and only if whenever $y_n \rightarrow y$ in Y then any closed set containing $f^{-1}(y_n)$ contains $f^{-1}(y)$ (what is the continuous equivalent of this?)
4. to gain intuition, show using the above that $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) \rightarrow x^2$ is not open. If you know some complex analysis, contrast this to $f : \mathbb{C} \rightarrow \mathbb{C}$ given by $f(z) = z^2$ which *is* an open map by the open mapping theorem for holomorphic functions.

Hence, openness gives inverse clustering information, whiled closedness gives inverse convergence information! The polynomial example is a bit of a facetious example as it is “solved” simply by extending the domain, and so any example involving analytic functions over $\mathbb{R}$ are facetious, however it is nonetheless instructive as a first example of using the above definitions.

Notice how an open this also gives an “almost” give a continuous inverse: we are guaranteed that we shall map (open) neighborhoods to (open) neighborhoods, and so if we are injective locally, we have a *continuous* inverse map.

When working with bounded linear operators, we shall see that the continuous linearity is strong to force the maps to be open, hence force the maps to preserve clustering in the pre-image:

Theorem 2.2.1: Open Mapping Theorem

Let X and Y be Banach spaces and $T \in B(X, Y)$ be a bounded linear operator with $T(X) = Y$. Then T is an open map (i.e. it is a surjective bicontinuous map)

Proof :

Since T is surjective:

$$Y = T(X) = \bigcup_n \overline{T(B_n(0))}$$

Then by corollary 2.2.1, not all closed sets $\overline{T(B_n(0))}$ have an empty interior. Thus, there is an n and a ball $B_\epsilon(y)$ such that

$$B_\epsilon(y) \subseteq \overline{T(B_n(0))}$$

Thus, $T(B_1(0))$ is dense in $B_{n^{-1}\epsilon}(y)$. Since

$$2B_{n^{-1}\epsilon}(0) \subseteq B_{n^{-1}\epsilon}(y) - B_{n^{-1}\epsilon}(y)(???)$$

$T(B_1(0))$ is also dense in $B_{n^{-1}\epsilon}(0)$. By Baire Category Theorem II, $B_\delta(0)$ is contained in $T(B_1(0))$ for every $\delta < n^{-1}\epsilon$. Since every open set $U \subseteq X$ is a union of balls, we conclude from linearity of T that $T(U)$ contain a neighborhood^a around each of its points, thus $T(U)$ is open, completing the proof.

^ain fact closed balls

Corollary 2.2.2: Bijective And Continuous

Let $T : X \rightarrow Y$ be a bijective bounded operator. Then it has a bounded inverse, that is T is in fact a Banach-space Isomorphism

Corollary 2.2.3: Equivalent Banach Spaces

Let X be a vector-space, and a Banach space under two norms $\|\cdot\|_a$ and $\|\cdot\|_b$. Then if $\|\cdot\|_a \leq \alpha \|\cdot\|_b$ for some $\alpha > 0$, then there is a $\beta > 0$ such that $\|\cdot\|_b \leq \beta \|\cdot\|_a$

Proof :

$\|\cdot\|_a$ is a surjective bounded operator, and hence open, and so has a bounded inverse

The next powerful result we get is a quick way of proving a function is bounded:

Theorem 2.2.2: Closed Graph Theorem

Let $T : X \rightarrow Y$ be an operator between Banach spaces. Then if the graph

$$\Gamma(T) = \{(x, y) \in X \times Y \mid T(x) = y\}$$

is closed in $X \times Y$, then T is bounded

Proof :

Using the ∞ -norm on $X \times Y$, and assume that $\Gamma(T)$ is closed in $X \times Y$. Then the projection map $\pi_1 : \Gamma(T) \rightarrow X$ $\pi(x, T(x)) = x$ is a norm decreasing bijective map, and hence has a bounded inverse π_1^{-1} . The other projection $\pi_2 : \Gamma(T) \rightarrow D$ given by $\pi(x, T(x)) = T(x)$ is also norm decreasing giving us π_2^{-1} . Hence, we have:

$$T = \pi_2 \circ \pi_1^{-1}$$

showing that T is bounded, completing the proof.

Finally, the following result is a way from passing from pointwise boundedness to uniform boundedness, only requiring the completeness of our domain.

Theorem 2.2.3: Principle Of Uniform Boundedness

Consider a family $\{T_\lambda \mid \lambda \in \Lambda\} \subseteq B(X, Y)$ where X and Y are Banach spaces. Then if each set $\{T_\lambda(x) \mid \lambda \in \Lambda, x \in X\}$ is bounded in Y , then

$$\{\|T_\lambda\| \mid \lambda \in \Lambda\}$$

is bounded

Proof :

Analysis now, p. 54

Corollary 2.2.4: Net Convergence

Let $(T_\lambda)_{\lambda \in \Lambda} \subseteq B(X, Y)$ be a net set $(T_\lambda(x))_{\lambda \in \Lambda}$ is bounded and converges in Y or every $x \in X$. Then there is a $T \in B(X, Y)$ such that $T_\lambda(x) \rightarrow T(x)$ for every $x \in X$.

Proof :

Analysis now p. 54

If our net is a sequence, then convergence of each sequence $(T_n(x))$ for all $x \in X$ automatically implies boundedness. It also suffices in the principle of uniform boundedness to know that each net is bounded and that the net is convergent for a dense set of elements.

2.3 Hahn-Banach Theorem

Let V be a Banach space over k (k being either $\mathbb{R}$ or $\mathbb{C}$). Since we are in the category of Banach spaces, we'll consider $X := \text{Hom}(V, k)$ to be the collection of all continuous linear functionals (i.e. bounded linear functionals). We will start by proving the Hahn-Banach extension theorem, which shows that there exists “enough” linear functions in X^* to make their study interesting.

Definition 2.3.1: Minkowski Functional

Let $m : V \rightarrow \mathbb{R}$ be a subadditive function $m(x+y) \leq m(x) + m(y)$ that is positive homogeneous $m(tx) = tm(x)$ for all $t \geq 0$ in $\mathbb{R}$. Then m is called a *Minkowski functional*.

Note that it is not a semi-norm since homogeneity only needs to hold for positive scalars. One can think of the subadditivity condition as being the “analysis-extension” of the additive condition; if it works for equality, it will work for the weaker case of “less-than”.

Lemma 2.3.1: Fundamental Lemma

If m is a Minkowski functional on a real vector-space V and φ is a functional on a linear subspace $Y \subseteq X$ that is dominated by m (so $\varphi(y) \leq m(y)$ for all $y \in Y$), then there exists a function $\tilde{\varphi}$ on X dominated by M such that $\tilde{\varphi}|_Y = \varphi$

Proof :

Analysis now p. 57. Also in Folland in the section on Linear Functionals.

Theorem 2.3.1: Hahn-Banach Extension theorem

If m is a seminorm on a vector space V and φ is a functional on a subspace $Y \subseteq V$ such that $|\varphi| \leq m$, then there is a functional $\tilde{\varphi}$ on X such that $|\tilde{\varphi}| \leq m$ and $\tilde{\varphi}|_Y = \varphi$

Proof :

Analysis now p. 57

Corollary 2.3.1: Hahn-Banach Corollary I

For every $x \neq 0$ in a normed space X , there is a $\varphi \in X^*$ with $\|\varphi\| = 1$ and $\varphi(x) = \|x\|$

Proof :

Define φ on $\mathbb{F}x$ by $\varphi(\alpha x) = \alpha\|x\|$. Then $\|\varphi\| = 1$. Applying Hahn-Banach on $Y = \mathbb{F}x$ and $m = \|\cdot\|$ yields the desired result.

Corollary 2.3.2: Hahn-Banach Corollary II

For every closed subspace $Y \subseteq X$ of a normed space X and $x \in X \setminus Y$, there is a $\varphi \in X^*$ with $\|\varphi\| = 1$, $\varphi|_Y = 0$, and $\varphi(x) = \inf \{\|x - y\| \mid y \in Y\}$

Proof :

Apply the first corollary on the normed space X/Y , then note that functionals in $(X/Y)^*$ may be regarded as elements in X^* that annihilate Y

Definition 2.3.2: Annihilator

For a subspace $Y \subseteq X$ of a normed space X , the *annihilator* of Y is defined as:

$$Y^\perp = \{\varphi \in X^* \mid \varphi(y) = 0, \forall y \in Y\}$$

Similarly, we define the annihilator of a subspace $Y^* \subseteq X^*$ to be

$$(Y^*)^\perp = \{x \in X \mid \varphi(x) = 0, \forall \varphi \in Y^*\}$$

It is immediate that $Y \subseteq (Y^\perp)^\perp$, and it follows by corollary 2.3.2 that in fact $Y = (Y^\perp)^\perp$ for every norm closed subspace $Y^* \subseteq X^*$.

(More here)

2.4 Fréchet Spaces: Limits of Banach Spaces

In analysis, it is very common to extend an object by taking a sequence of them and defining a sort of convergence. Fréchet spaces follow exactly this principle, where a topological vector-space shall be defined using a countable family of semi-norms. Semi-norms are used as it is often the case that the space is “too big” for the function $\|\cdot\| : X \rightarrow \mathbb{R}$ to be faithful ($\|x\| = 0$ iff $x = 0$). Most types of convergences are accomplished by a family of semi-norms. For example, convergence on compact sets, pointwise convergence, uniform convergence, and smooth convergence are all convergence in Fréchet spaces. In analysis, there will be many spaces that will Fréchet spaces, notably when you encounter distributions and require to define a topology on the dual space. Hence, a moment to get a better understanding of them is worth it.

Theorem 2.4.1: Semi-Norms Generate TVS

Let $\{p_\alpha\}_{\alpha \in A}$ be a family of norms on a vector space V . If $x \in V$, $\alpha \in A$, and $\epsilon > 0$, let

$$U_{x,\alpha,\epsilon} = \{y \in V \mid p_\alpha(y - x) < \epsilon\}$$

and let $\mathcal{T}$ be the topology generated by the sets $U_{x,\alpha,\epsilon}$. Then:

1. For each $x \in V$ the finite intersections of the sets $U_{x,\alpha,\epsilon}$ ($\alpha \in A, \epsilon > 0$) form a neighborhood base at x
2. if $\langle x_i \rangle_{i \in I}$ is a net in V , then $x_i \rightarrow x$ if and only if $p_\alpha(x_i - x) \rightarrow 0$ for all $\alpha \in A$.
3. $(V, \mathcal{T})$ is a locally convex topological vector-space

Proof :

Folland p. 166

Definition 2.4.1: Fréchet Space

A complete Hausdorff topological vector-space whose topology is defined by a countable family of seminorms is called a *Fréchet space*.

(theorems on how to characterize continuous maps and when a space is Hausdorff, and also if it is Hausdorff, it is equivalent to define a translation-invariant metric)

Hilbert Spaces

(Another intuition to incorporate: Since Hilbert spaces are self-dual due to Riesz representation theorem, they have a very simple duality theory. Think of how C^∞ does is not self-dual, and that's why we have “non-trivial” distributions, or how $(L^p)^*$ has a very easy isomorphism to L^q . In the case of a Hilbert space H , we have $H^* \cong H$, and so any extra information we might try to get from a dual (ex. notion of size, boundedness, oscillation, regularity, decay, etc.) can be represented by an element of the Hilbert space! This makes the dual theory easier to study!!)

(Another important property of Hilbert spaces is that we can always find a dense set that is *linear*. That is, any element $x \in H$ is equal to a convergent sequence $x = \sum_i^\infty x_i$. This is in a sense the best type of dense set!)

(Some geometric properties that are lost in Banach spaces is that we may have unit balls with corners, or may have closed convex sets with no elements of minimal norm. There is also no notion of “perpendicularity” and no good notion of a basis, at best having dense sets)

When dealing with Euclidean Geometry, we have many useful conceptions such as the triangle inequality, the notion of perpendicularity, the idea notion of “area” and the translation invariance of said area. We generalized these notion to vector spaces over $\mathbb{R}$ or $\mathbb{C}$, which preserved many of these intuition's in the finite-dimensional case. However, moving up to the infinite dimensional case, these intuitions are no longer necessarily preserved (examples here?). Furthermore, even in the finite dimensional case over $\mathbb{R}$ and $\mathbb{C}$, without a topology that permits the notion of completeness, we cannot do many of the constructions we've been doing thus far that are dependent upon completion (essentially, everything we have been working with).

The goal of this chapter is to be able to talk about infinite dimensional vector-spaces which still have some geometric notions, as well as some topological notions. Perhaps the most common such example would be function spaces (some of which we will explore in this section), where many are infinite dimensional. It would be nice if our notion of geometry also applied to them, and so the goal of this section is to see whether they do.

Note that the notion of a metric and a norm is not sufficiently strong enough to capture the geometrical ideas we will present: the inner product really is necessary to induce these ideas, as we shall soon see.

3.1 Basic definition and properties

Definition 3.1.1: Inner Product Space

Let H be a complex vector space with an inner product $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}$ where

1. $\langle x, y \rangle = \overline{\langle y, x \rangle}$
2. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
3. $\langle ax, y \rangle = a\langle x, y \rangle, a \in \mathbb{C}$
4. $\langle x, x \rangle \in (0, \infty)$ for all nonzero $x \in H$
5. $\langle x, x \rangle = 0$ if and only if $x = 0$

We'll define $\|x\| = \sqrt{\langle x, x \rangle}$. This is naturally a norm since:

1. $\|x\| = \sqrt{\langle x, x \rangle} = 0$ if and only if $x = 0$
2. $\|ax\| = \sqrt{\langle ax, ax \rangle} = \sqrt{a\bar{a}\langle x, x \rangle} = \sqrt{a^2\langle x, x \rangle} = |a|\sqrt{\langle x, x \rangle} = |a|\|x\|$

Proving the triangle inequality is a bit more difficult: If we square both sides, we it is equivalent to show:

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2$$

expanding out the left hand side, we get:

$$\|x + y\|^2 = \|x\|^2 + 2\operatorname{Re}\langle x, y \rangle + \|y\|^2$$

so we have to show that $\operatorname{Re}\langle x, y \rangle \leq \|x\|\|y\|$. In fact, a bit more is true: $|\langle x, y \rangle| \leq \|x\|\|y\|$. Though we are using this inequality to prove the triangle inequality, notice that this in fact gives us a general bound on the value of the inner product! This inequality is one of the central tools used in the study of Hilbert spaces:

Theorem 3.1.1: Cauchy-Schwartz Inequality

$$|\langle x, y \rangle| \leq \|x\|\|y\|$$

Proof :

If $\|y\| = 0$ then $y = 0$, and then $|\langle x, 0 \rangle| = 0$, so assume $\|y\| \neq 0$. Then for all $t \in \mathbb{R}$

$$\begin{aligned}
 0 &\leq \langle x - ty, x - ty \rangle \\
 &= \|x\|^2 + 2t \operatorname{Re} \langle x, y \rangle + t^2 \|y\|^2 \\
 &= \|x\|^2 + \frac{2 \operatorname{Re} \langle x, y \rangle^2}{\|y\|^2} + \frac{\operatorname{Re} \langle x, y \rangle^2}{\|y\|^2} \quad t = \frac{\operatorname{Re} \langle x, y \rangle}{\|y\|^2} \\
 &\Rightarrow |\operatorname{Re} \langle x, y \rangle| \leq \|x\| \|y\|
 \end{aligned}$$

To finish the proof, pick $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ such that

$$|\operatorname{Re}(\alpha \langle x, y \rangle)| = |\langle x, y \rangle|$$

Notice this does not break the inequality, and so we can repeat the entire process but with α taken into account with t , and so:

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

as we sought to show

With this, the rest of the proof that $\|\cdot\|$ induced by $\langle \cdot, \cdot \rangle$ is a norm follows

Theorem 3.1.2: Triangle Inequality For Induced Norm

$$\|x + y\| \leq \|x\| + \|y\|$$

Proof :

$$\begin{aligned}
 \|x + y\|^2 &= \langle x + y, x + y \rangle \\
 &\leq \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\
 &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \\
 &= (\|x\| + \|y\|)^2
 \end{aligned}$$

The Cauchy-Schwartz inequality is the tool we use to bring in geometry as we know it into Hilbert space. For example, in real inner product spaces, we will define the angle between two vectors to be θ where θ satisfies:

$$\langle u, v \rangle = \cos(\theta) \|u\| \|v\|$$

Many other cool properties of the Cauchy-Schwartz inequality can be found here: (commented)

Now, notice we can define a metric by doing $\rho(x, y) = \|x - y\|$:

1. $\rho(x, x) = \|x - x\| = 0$ if and only if $x - x = 0$ so $x = x$
2. $\rho(x, y) = \|x - y\| \geq 0$
3. $\rho(x, y) = \|x - y\| = \|x - y + y - z\| \leq \|x - y\| + \|y - z\| = \rho(x, y) = \rho(y, z)$

Thus, we can define a natural topology on any pre-hilbert space. If that topology is complete, we give it a name:

Definition 3.1.2: Hilbert Space

Let H be an inner product space. Then H is a Hilbert space if it is *complete* with respect to the induced metric.

Example 3.1: Hilbert Spaces

1. $L^2(\mu)$ for a complex measure μ is a Hilbert space with the inner product being:

$$\langle f, g \rangle_{L^2} = \int_X f(x) \overline{g(x)} d\mu(x)$$

each axiom is simple to verify:

2. $\mathbb{R}^n$ is a Hilbert space with the inner product being:

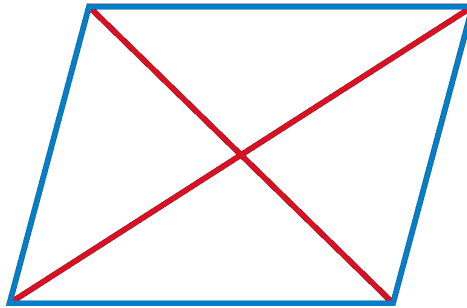
$$\langle x, y \rangle = \sum_i x_i y_i$$

Hilbert spaces are very euclidean in nature, for example:

Theorem 3.1.3: Parralelogram

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Visually, this means that the “area” of a parralelogram is can be measured in two ways:



Proof :

$$\begin{aligned}
 \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\
 &= 2\langle x, x \rangle + \langle x, y \rangle - \langle x, y \rangle + \langle y, x \rangle - \langle y, x \rangle + 2\langle y, y \rangle \\
 &= 2\langle x, x \rangle + 2\langle y, y \rangle \\
 &= 2\|x\|^2 + 2\|y\|^2
 \end{aligned}$$

It turns out that a norm satisfying the parallelogram law will in fact define an inner product. This shows how norms define geometries that are more general than euclidean:

Example 3.2: Exploring Parralelogram Law

1. (The polarization Identity) For any $x, y \in \mathcal{H}$,

$$\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2)$$

Proof. This is simply a matter of computation:

$$\begin{aligned}
 &\frac{1}{4}(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2) \\
 &= \frac{1}{4}(\langle x + y, x + y \rangle - \langle x - y, x - y \rangle + i\langle x + iy, x + iy \rangle - i\langle x - iy, x - iy \rangle) \\
 &= \frac{1}{4}(\langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle - (\langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle) \\
 &\quad + i(\langle x, x \rangle + \langle x, iy \rangle + \langle iy, x \rangle + \langle iy, iy \rangle - (\langle x, x \rangle - \langle x, iy \rangle - \langle iy, x \rangle + \langle iy, iy \rangle)) \\
 &= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle + i(2\langle x, iy \rangle + 2\langle iy, x \rangle)) \\
 &= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle + i(-i)2\langle x, y \rangle + 2ii\langle y, x \rangle) \\
 &= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle + 2\langle x, y \rangle - 2\langle y, x \rangle) \\
 &= \frac{1}{4}4\langle x, y \rangle \\
 &= \langle x, y \rangle
 \end{aligned}$$

□

2. If $\mathcal{H}'$ is another Hilbert space, a linear map from $\mathcal{H}$ to $\mathcal{H}'$ is unitary if and only if it is isometric and surjective

Proof. Let U be unitary so that $\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$ for any $x, y \in \mathcal{H}$. Letting $x = y$ gives us $\|Ux\|_2 = \|x\|_1$, giving us that it's an isometry. Since U is invertible, it is surjective, and so U is a surjective isometry.

Conversely, let U be a surjective isometry. We need to show its invertible (it suffices to show it's injective), an isometry, and that U and U^{-1} are bounded. Clearly, U is bounded since $\|U(x)\| \leq 1\|x\|$. For injectivity, let $Ux = Uy$, so $Ux - Uy = 0$. Taking the norm on both sides, we get:

$$0 = \|0\|_2 = \|Ux - Uy\|_2 = \|U(x - y)\|_2 = \|x - y\|_1$$

where the last equality is due to U being an isometry. Since $\|z\| = 0$ if and only if $z = 0$, we have $x - y = 0$, i.e. $x = y$, proving that U is injective. Along with surjectivity, that makes U invertible. With this, we can also show that U^{-1} is bounded: for any $y \in \mathcal{H}'$, there exists an $x \in \mathcal{H}$ such that $U(x) = y$. Thus:

$$\|U^{-1}(y)\| = \|x\| \stackrel{!}{=} \|U(x)\| = \|y\|$$

where $\stackrel{!}{=}$ comes from U being an isometry, showing that $\|U^{-1}(y)\| \leq 1 \cdot \|y\|$, making it also bounded.

Finally, to show that $\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$, we will use part (a):

$$\begin{aligned} \langle x, y \rangle &= \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2) \\ &\stackrel{!}{=} \frac{1}{4}(\|U(x + y)\|^2 - \|U(x - y)\|^2 + i\|U(x + iy)\|^2 - i\|U(x - iy)\|^2) \\ &= \frac{1}{4}(\|U(x) + U(y)\|^2 - \|U(x) - U(y)\|^2 + i\|U(x) + iU(y)\|^2 - i\|U(x) - iU(y)\|^2) \\ &= \langle U(x), U(y) \rangle \end{aligned}$$

where $\stackrel{!}{=}$ comes from U being an isometry. Thus:

$$\langle Ux, Uy \rangle = \langle x, y \rangle$$

As we sought to show. □

Proposition 3.1.1: Convergence In Inner Product

Let $x_n \rightarrow x$ and $y_n \rightarrow y$. Then $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$

Proof :

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &= |\langle x_n - x, y_n \rangle + \langle x, y_n - y \rangle| \\ &= \|x_n - x\| \|y_n\| + \|x\| \|y_n - y\| \end{aligned}$$

which tends to zero since $x_n, y_n \rightarrow 0$.

When it comes to the right notion of morphism for Hilbert spaces, this is actually not as straight forward as it may seem. To start with a special case, it is straightforward to imagine an isomorphism:

Definition 3.1.3: Unitary Map

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two hilbert spaces with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$, and let $U : \mathcal{H}_1 \rightarrow \mathcal{H}_2$. Then U is called a *unitary map* if it is an invertible linear map that perserves inner products:

$$\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$$

The appropriate notion of a morphism that matches this is not so easy. We may result to the fact

that all Hilbert spaces are Banach spaces and use continuous linear operators, however such operators do not capture the inner product structure (it cannot be recovered using this information). There is a way to recover this information if we instead upgrade our category with an involution, this is known as a dagger category:

$$(-)^\dagger : \text{Hom}(a, b) \rightarrow \text{Hom}(b, a)$$

this is a morphism between homsets that satisfy certain properties given by adjoint (which we'll see in the next section). Then we recover an inner product and unitary maps. We won't go down the road of looking at dagger categories, but it is insightful to think about what information is useful in these categories.

Note that by taking $x = y$, we get that $\|x\|_1 = \|Ux\|_2$, i.e. it is an isometry. In fact, the converse is true too: every surjective isometry is unitary. We can define the operator norm on $\|T\|$ differently using the inner product:

$$\|T\| = \sup_{\|x\|=1, \|y\|=1} |\langle T(x), y \rangle|$$

the two are shown to be equivalent with not too much work:

Proof. For $\geq$, using Cauchy-Swartz we get:

$$\sup_{\|x\|=1, \|y\|=1} |\langle T(x), y \rangle| \leq \|T(x)\| \leq \|T\|$$

For the other direction

$$\begin{aligned} \|T\| &= \sup_{\|x\|=1} \|T(x)\| \\ &= \sup_{\|x\|=1, T(x) \neq 0} \frac{\langle T(x), T(x) \rangle}{\|T(x)\|} \\ &\leq \sup_{\|x\|=\|y\|=1} |\langle T(x), y \rangle| \end{aligned}$$

□

Subspaces

Just like for Banach spaces, closed subspaces of Hilbert spaces are Hilbert spaces. As we saw in section 2.3, we often want to separate an operator by how it *annihilates* a subspace. For Hilbert spaces, such subspaces are really easy to find:

Definition 3.1.4: Orthogonal

We say that x is *orthogonal* to y if $\langle x, y \rangle = 0$. We will write $x \perp y$

If two vectors are orthogonal, the triangle inequality becomes instant:

Theorem 3.1.4: Pythagoras

$$x \perp y \Rightarrow \|x + y\|^2 = \|x\|^2 + \|y\|^2$$

Proof :

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + \|y\|^2$$

We might wonder if we take $E \subseteq H$, can we find all vectors in H that are somehow “dual” to E ? Define it's *orthogonal compliment* to be:

$$E^\perp = \{x \in H \mid \langle x, y \rangle = 0 \ \forall y \in E\}$$

Notice that E is *always* a closed subspace:

Proof. The fact it's a subspace follows from linearity of the second component.

for closed, suppose there is a sequence $\{x_n\} \subseteq E^\perp$ where $x_n \rightarrow x \in H$. Then:

$$|\langle x, y \rangle| \leq |\langle x - x_n, y \rangle| \leq \|x - x_n\| \|y\| \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

□

We can also find closed (and not closed) subspace of hilbert spaces:

Example 3.3: Subspaces Examples

1. If $\mathcal{H}$ is a finite-dimensional Hilbert space, then all subspace are closed
2. If $\mathcal{H}$ is infinite-dimensional, consider $\mathcal{H} = L^1([0, 1])$ with the L^2 inner product. and take $P([0, 1]) \subseteq L^1([0, 1])$ to be the subspace of all polynomial functions. This is certainly not equal to $L^1([0, 1])$, however polynomial functions are dense in $L^1([0, 1])$ (they can arbitrarily approximate continuous bump characteristic functions, and hence simple functions). However, their closure is $L^1([0, 1])$, showing that $P([0, 1])$ cannot be closed.
3. Assume $E \subseteq \mathcal{H}$ is an open subspace. Then since $0 \in E$, there exists a $r > 0$ such that

$$B_r(0) \subseteq E$$

However, for any nonzero element $x \in \mathcal{H}$,

$$\frac{r}{2\|x\|}x \in B_r(0) \subseteq E$$

Since E is closed under scalar multiplication, this shows that $x \in E$, so $\mathcal{H} \subseteq E$, so $\mathcal{H} = E$, showing that the only open subspace of $\mathcal{H}$ is $\mathcal{H}$ itself.

Thus, we are interested in closed subspace since are closed under limits, meaning we can use our tools from analysis to get the following decomposition result:

Theorem 3.1.5: Hilbert Space Decomposition

Let $E \subseteq \mathcal{H}$ be a closed subspace. Then:

$$\mathcal{H} = E \oplus E^\perp$$

In particular, for any $x \in H$, there is a unique $y \in E$ and $z \in E^\perp$ such that $x = y + z$.

Proof :

Let $x \in \mathcal{H}$, and let $\delta = \inf \{\|x - y\| = \rho(x, y) \mid y \in E\}$ (which exists by completeness of $\mathbb{R}^n$ and $\mathcal{H}$). let $\{y_n\} \subseteq E$ be a sequence such that $\|x - y_n\| \rightarrow \delta$. We will show that $\{y_n\}$ is a cauchy sequence, and then find a limit point for it. By the parralelogram law:

$$2(\|x - y_n\| + \|x - y_m\|) = \|y_n - y_m\|^2 + \|(y_n + y_m) - 2x\|^2$$

Since $y_n + y_m \in E$, for sufficinetly large N so that $n, m > N$:

$$\begin{aligned} \|y_n - y_m\|^2 &= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\left\|\frac{1}{2}(y_n + y_m) - x\right\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\delta^2 \end{aligned}$$

and as $n, m \rightarrow \infty$, this sequence goes to $4\delta^2 - 4\delta^2 = 0$, and so $\|y_n - y_m\| \rightarrow 0$, meaning $\{y_i\}_i$ is a cauchy sequence. Take $y = \lim y_n$. Then since E is closed, $y \in E$. By construction $\|x - y\| = \delta$. Now let $z = x - y$.

I claim that $z \in E^\perp$, that is $\langle z, u \rangle = 0$ for all $u \in E$. So, consider $\langle u, z \rangle$ for any $u \in E$. If $\langle u, z \rangle$ is complex, multiply it by a [complex] unit-length scalar to make it real ($\langle u, z \rangle = 0$ if and only if $a\langle u, z \rangle = 0$). With the fact that it's real, we will do a sneaky trick where we take advantage of optimization of differentiable functions. In particular, let

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f(t) = \|z + tu\|^2 = \|z\|^2 + 2t\langle z, u \rangle + \|u\|^2$$

f is certainly a differentiable function, being a polynomial with a square-root on $\mathbb{R}$. Furthermore, $z = x - y$ is the minimal value of $\|x - y\|$, and so the minimal value of $\|x - y\|^2$. It follows that $z + ut$ is not minimal, and so this equation is has a minimum when $t = 0$. Thus, we get $0 = f'(0) = 2\langle u, z \rangle$, implying that $\langle u, z \rangle = 0$. Since u was arbitrary, we get that $z \in E^\perp$.

To show that z is the minimal distance from x , let's say there was another $z' \in E^\perp$ that satisfied the equation. Then by the Pythagorean theorem:

$$\begin{aligned} \|x - z'\| &= \|(x - z) + (z - z')\|^2 \\ &= \|x - z\|^2 + \|z - z'\|^2 \\ &\geq \|x - z\|^2 \end{aligned} \quad x - y \in E, z - z' \in E^\perp$$

So equality holds if and only if $z = z'$, meaning z is of minimal distance to x . A similar reasoning shows the same for y .

To show uniqueness, let's say $y + z = x = y' + z'$. Then $y - y' = z - z' \in E \cap E^\perp = \{0\}$, so $y - y' = z - z' = 0$, which can be used to show that $y = y'$ and $z = z'$, completing the proof.

Corollary 3.1.1: Double Dual Property

For each subset $Y \subseteq H$, the smallest closed subspace of H containing Y is $(Y^\perp)^\perp$. In particular, if Y is a subspace of H , then $\overline{X} = (X^\perp)^\perp$

Proof :
exercise

We said that orthogonal vector are somehow dual to each other. However, there is the more classical notion of “dual” vector spaces. Let $\mathcal{H}^*$ be set of all linear functions from $\mathcal{H}$ to the underlying field (usually $\mathbb{C}$, but sometimes $\mathbb{R}$); such linear functions are called functionals. It turns out that all linear functionals can be written in the form $f_y(x) = \langle x, y \rangle$, and we can define a norm on $\mathcal{H}^*$ such that $\|f_y\| = \|y\|$. Thus, $y \mapsto f_y$ will define a conjugate-linear isometry from $\mathcal{H}$ to $\mathcal{H}^*$. The following theorem shows tha this map is surjective:

Theorem 3.1.6: Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)

Let $f \in \mathcal{H}^*$. Then there exists a $y \in \mathcal{H}$ such that $f(x) = \langle x, y \rangle$ for all $x \in X$.

Proof :

It easy easy to check uniqueness: if $\langle x, y \rangle = \langle x, y' \rangle$ for all x , then let $x = y - y'$. Manipulating the equation, we get that $\|y - y'\| = 0$, so $y = y'$.

If f is the zero functional, then take $y = 0$. If otherwise, then let $\mathcal{M} = \{x \mid f(x) = 0\}$ be the kernel of f . It is clearly a subspace, and any squnce will also necesarily converge within $\mathcal{M}$, and so it is also closed. Furthermore, $\mathcal{M}$ is a proper subspace, since $f \neq 0$. Thus, $\mathcal{M}^\perp \neq \{0\}$ by the previous proof.

Now, pick any $z \in \mathcal{M}^\perp$ with $\|z\| = 1$. Then if we set $u = f(x)z - f(z)x$, we have that $u \in \mathcal{M}$. Hence:

$$0 = \langle u, z \rangle = f(x)\|z\|^2 - f(z)\langle x, z \rangle = f(x) - \langle x, \overline{f(z)}z \rangle$$

re-arrangine and setting $y = \overline{f(z)}z$, we get

$$f(x) = \langle y, z \rangle$$

as we sought to show.

The uniqueness makes this map to be injective, and by linearity of f and the first component of the inner product makes it is an isomorphism. Furthermore, since $\|f_y\| = \|y\|$, limits are also preserved, and so this in fact defines an isomorphism between $\mathcal{H} \cong \mathcal{H}^*$. Notice how this differs from vector-spaces where there is no natural isomorphism between V and V^* (there is one between V and V^{**}).

As we’ve done with banach spaces, we should define the topology on the dual of H . By the Riesz representation theorem, we see that any $f \in H^*$ is simply $\langle \cdot, y \rangle$ for appropriate $y \in H$. Hence, the weak topology on H is the initial topology corresponding to the functionals $x \mapsto \langle x, y \rangle$.

Note that it is important that the space is *complete*. The following shows what can happen otherwise:

Example 3.4: Functional Without Riesz Representation

Let $V = \mathbb{C}[x]$ where we treat any $f \in \mathbb{C}[x]$ as a function $f : [0, 1] \rightarrow \mathbb{C}$. Then:

$$\langle f, g \rangle = \int_0^1 f(t)\overline{g(t)}dt$$

Note that V is not complete: Letting $V \subseteq C([0, 1])$ we see there is a Cauchy sequence in V that converges to $|x - 0.5|$ in $C([0, 1])$. Let's say that Riesz Representation theorem still holds, so that for $L_z \in V^*$ where $L(f) = f(z)$, there exists a $g \in V$ such that

$$L(f) = \langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt$$

Then

$$f(z) = \int_0^1 f(t) \overline{g(t)} dt$$

For any $f \in V$. Define $h(z) = x - z$ so that $hf \in V$ and $(hf)(z) = h(z)f(z) = 0$. Hence:

$$0 = \int_0^1 h(t) f(t) \overline{g(t)} dt$$

If $f = \overline{h}g \in V$ then

$$0 = \int_0^1 |h(t)|^2 |g(t)|^2 dt$$

since $h \neq 0$, it must be that $g = 0$ a.e. But then $\langle f, g \rangle = 0$ a.e., so $L_z(f) = 0$ almost everywhere. But that's a contradiction, as certainly L_z is not 0 a.e.

However, notice that if we take the completion, we will have enough elements so that L_z does indeed have a Riesz Representation.

Orthogonal Basis

In this section, we will show how Hilbert spaces have much nicer dense sets than Banach spaces. In particular, we shall see that we will be able to upgrade the notion of a basis more readily to Hilbert spaces. Since we are studying infinite dimensional vector-spaces, we often care more about “countable linear combination” rather than “finite linear combination” as we have for vector-spaces. To distinguish the two, we will call *Hamel basis* a set of linearly independent elements for which all *finite linear combinations* can represent every element of the vector space. We will call a *Schauder basis* a set of linearly independent elements for which all *countable linear combinations* can represent every element of the vector space. If every element in a Schauder basis has norm 1 and is orthogonal, we call it an *orthonormal basis*. Since we have an inner-product that is complete, it would be nicest if this vector-space has an orthonormal basis. We strive to this end in the following build-up:

Theorem 3.1.7: Bessel's Inequality

If $\{u_\alpha\}_{\alpha \in A}$ is an orthonormal set in $\mathcal{H}$, then for any $x \in \mathcal{H}$:

$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$$

As a consequence, $\{\alpha \mid \langle x, u_\alpha \rangle \neq 0\}$ is countable.

Proof :

It suffices to show that $\sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$ for any finite subset $F \subseteq A$, since if it fails for some countable amount, it will fail for some finite amount. Then we see that:

$$\begin{aligned}
 0 &\leq \left\| x - \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\|^2 \\
 &= \|x\|^2 - 2 \operatorname{Re} \left\langle x, \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\rangle + \left\| \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\|^2 \\
 &= \|x\|^2 - 2 \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 + \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 && \text{Pythagorean Theorem} \\
 &= \|x\|^2 - \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2
 \end{aligned}$$

and manipulating the final equation we get:

$$\sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$$

as we sought to show

This in particular is important for uncountable sets. In the following we establish when equality holds, and thus when an orthonormal set is an orthonormal basis

Theorem 3.1.8: Orthonormal Basis Condition

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set in $\mathcal{H}$. Then the following are equivalent:

1. (Completeness) If $\langle x, u_\alpha \rangle = 0$ for all α , then $x = 0$
2. (Parseval's Identity) $\|x\|^2 = \sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2$ for all $x \in \mathcal{H}$
3. For each $x \in \mathcal{H}$, $x = \sum_{\alpha \in A} \langle x, u_\alpha \rangle u_\alpha$ where the right hand side sum has only countably many nonzero terms and converges in the norm topology no matter the order of the terms (so converges absolutely?)

Proof :

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Definition 3.1.5: Orthonormal Basis

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set. Then if the set satisfies any of the 3 conditions in theorem 3.1.8, it is called an *orthonormal basis*.

Theorem 3.1.9: Hilbert Space Basis

Every Hilbert space has a basis

Proof :

Application of Zorn's lemma: The collection of ordered orthonormal sets ordered by inclusion has a maximal element, and maximality is equivalent to the first property of theorem 3.1.8.

Now that we have established a basis for Hilbert spaces, we will separate them into 3 categories: Those with finite, countable, and uncountable basis. In most cases, we will be working with countable basis due to the following property:

Proposition 3.1.2: Seperable Hilbert Spaces

A Hilbert space $\mathcal{H}$ is seperable if and only if it has a countable orthonormal basis, in which case every orthonormal basis for $\mathcal{H}$ is countable.

Here, we are using the word seperable to mean countable dense subset.

Proof :

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With the definition of an isomorphism, we can see that all Hilbert spaces are isomorphic to $\ell^2(A)$ given basis A :

Theorem 3.1.10: Classifying Hilbert Spaces

Let $\{e_\alpha\}_{\alpha \in A}$ be an orthonormal basis for $\mathcal{H}$. Then the correspondence

$$x \mapsto \hat{x} \quad \hat{x}(\alpha) = \langle x, u_\alpha \rangle$$

is a unitary map from $\mathcal{H}$ to $\ell^2(A)$.

Proof :

We'll show it's a surjective isometry, which is equivalent to it being unitary.

First off, the map $x \rightarrow \hat{x}$ is linear since the inner product is linear in each component, and it is an isometry by Parseval identity: $\|x\|^2 = \sum |\hat{x}(\alpha)|^2$.

For surjectivity, let $f \in \ell^2(A)$, so $\sum_{\alpha \in A} |f(\alpha)|^2 < \infty$, so the Pythagorean shows that all partial sumns of the series $\sum f(\alpha)u_\alpha$ are Cauchy ^a, so $x = \sum_{\alpha} f(\alpha)u_\alpha$ exists in $\mathcal{H}$ and $\hat{x} = f$.

^aNote that only countably many terms are nonzero

3.2 Operators on Hilbert Spaces

In this section, we try to generalize many notion's we've seen before, including:

1. adjoints, self-adjoint, and normal operators
2. sign and absolute value and when can we decompose a function into this form
3. diagonalizable
4. polar decomposition (think of how for every $z \in \mathbb{C}$, $z = \arg(z)|z|$)

Adjoint

Hilbert spaces have very simple dual structure, since $\mathbb{H} \cong \mathbb{H}^*$ in a very natural way with the inner product of $\mathbb{H}$. Thus, the problem of finding the structure of dual spaces is fully finished for hilbert spaces. Due to this very simple dual structure, we will be able to properly define an “opposite” operator given any operator $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$. In essence, there is an interesting symmetry in the structure of any map: Given a map $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ between two hilbert spaces, there is a natural dual map $T^* : \mathbb{H}_2^* \rightarrow \mathbb{H}_1^*$. Hence, we have

$$(-)^* : \text{Hom}(\mathbb{H}_1, \mathbb{H}_2) \rightarrow \text{Hom}(\mathbb{H}_2^*, \mathbb{H}_1^*)$$

and since in Hilbert spaces $\mathbb{H}_i \cong \mathbb{H}_i^*$, we may envision this map as “flipping” the direction of arrows in a natural way:

$$(-)^* : \text{Hom}(\mathbb{H}_1, \mathbb{H}_2) \rightarrow \text{Hom}(\mathbb{H}_2, \mathbb{H}_1)$$

Not only does this map exist, but it is an isometric isomorphism (in **Ban**)! We will start with the special case of $\text{Hom}(\mathbb{H}, k) = \mathbb{H}^*$. We will find that $(-)^*$ is a map

$$(-)^* : \text{Hom}(\mathbb{H}, \mathbb{H}) \rightarrow \text{Hom}(\mathbb{H}, \mathbb{H})$$

The following shows how we can define such a dual and some consequences:

Lemma 3.2.1: Endomorphism and And Inner Products

Let $\mathbb{H}$ be a complex hilbert space. Then there is a bijective isometric correspondence between $\text{End}_k(\mathbb{H})$ and bounded sesquilinear forms on $\mathbb{H}$ given by the map $T \rightarrow B_T$ where

$$B_T(x, y) = \langle x, T(y) \rangle$$

Proof :

Let $T \in \text{End}_k(\mathbb{H})$. Then certainly B_T is a sesquilinear form on $\mathbb{H}$ bounded by $\|T\|$ since

$$\begin{aligned} \|B_T\| &= \sup \{ |B_T(x, y)| \mid \|x\| \leq 1, \|y\| \leq 1 \} \\ &= \sup \{ |\langle x, T(y) \rangle| \mid \|x\| \leq 1, \|y\| \leq 1 \} \\ &\leq \|T\| \end{aligned}$$

On the other hand

$$\begin{aligned} \|T(x)\|^2 &= \langle T(x), T(x) \rangle \\ &= B_T(T(x), x) \\ &\leq \|B_T\| \|T(x)\| \|x\| \\ &\leq \|B_T\| \|T\| \|x\|^2 \end{aligned}$$

showing that $\|T\| \leq \|B_T\|$, and so $\|T\| = \|B_T\|$.

Theorem 3.2.1: Existence Of Adjoint

Let $T \in \text{End}_k(\mathbb{H})$. Then there exists a unique T^* in $\text{End}_k(\mathbb{H})$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

for all $x, y \in \mathbb{H}$. The map $T \mapsto T^*$ is conjugate linear, contravariant with respect to multiplication, an isometry of $\text{End}_k(\mathbb{H})$ onto itself, an involution, and satisfies

$$\|T^*T\| = \|T\|^2$$

Proof :

Fix $T \in \text{End}_k(\mathbb{H})$, and let $B_T(x, y)$ be a continuous sesquilinear form defined as $(x, y) \mapsto \langle T(x), y \rangle$. Then by lemma 3.2.1, there exists a $T^* \in \text{End}_k(\mathbb{H})$ such that $T^* \rightarrow B_T$. In particular

$$B_T(x, y) = \langle x, T^*(y) \rangle$$

Thus,

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

giving us existence of such an operator. We now establish all the properties named in the theorem. By conjugate symmetry

$$\begin{aligned} \langle x, T^{**}(y) \rangle &= \langle T^*(x), y \rangle \\ &= \overline{\langle y, T^*(x) \rangle} \\ &= \overline{\langle T(y), x \rangle} \\ &= \langle x, T(y) \rangle \end{aligned}$$

Thus, T^{**} and T correspond to the same sesquilinear form, thus $T^{**} = T$. This also shows that $T \mapsto T^*$ is an involution. Furthermore:

$$\langle x(ST)^*y \rangle = \langle S(T(x)), y \rangle = \langle T(x), S^*(y) \rangle = \langle x, T^*(S^*(y)) \rangle$$

thus $(S \circ T)^* = T^* \circ S^*$ showing $(-)^*$ is a contravariant endofunctor. Since the operator norm is submultiplicative, we have $\|T^*T\| \leq \|T\|\|T^*\|$. On the other hand, applying $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ to T^* we get

$$\|T(x)\|^2 = \langle T(x), T(x) \rangle = \langle T^*(T(x)), x \rangle = \|T^*T\| \|x\|^2$$

showing that $\|T\|^2 \leq \|T^*T\|$

. Combining these we get $\|T\| \leq \|T^*\|$, hence $\|T\| = \|T^*\|$ (since $T^{**} = T$). This finishes the proof.

Note that if $\mathbb{H}$ is a finite dimensional complex Hilbert space, then each operator T has a matrix representation, say (α_{ij}) . Then $T^* = (\overline{\alpha_{ji}})$, i.e. $T^* = \overline{T}^t$ where T^t is the transpose given the matrix representation.

Definition 3.2.1: Adjoint And Self-Adjoint

Let $T \in B(\mathbb{H})$ be a continuous linear operator between Hilbert spaces. Then T^* is called the *adjoint* of T . If $T = T^*$, then T is said to be *self adjoint* (or *Hermitian*).

Proposition 3.2.1: Adjoint Properties I

Let $T \in \text{End}_k(\mathbb{H})$. Then

$$\ker(T^*) = (T(\mathbb{H}))^\perp$$

Proof :

Since $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ we have $y \in \ker(T^*)$ when $y \in (T(\mathbb{H}))^\perp$. Conversely, if $y \in (T(\mathbb{H}))^\perp$, then $T^*(y) \in \mathbb{H}^\perp = \{0\}$.

For ease of reference, we shall list some of the results we have proven in the theorem:

Proposition 3.2.2: Adjoint Properties II

Let $T, S \in B(\mathbb{H})$. Then:

1. $(T + S)^* = T^* + S^*$
2. $(cT)^* = \bar{c}T^*$
3. $(TS)^* = S^*T^*$
4. $(T^*)^* = T$
5. $I^* = I$

Proof :

If you're not convinced of any of these, double check the theorem or try proving it again.

Proposition 3.2.3: Adjoint Properties III

Let $T \in \text{End}_k(\mathbb{H})$. Then the following are equivalent:

1. T is invertible (so $T^{-1} \in \text{End}_k(\mathbb{H})$)
2. T^* is invertible
3. T and T^* are bounded away from zero.
4. T and T^* are injective, and $T(\mathbb{H})$ is closed
5. T is injective and $T(\mathbb{H}) = \mathbb{H}$

Proof :

Since T is invertible, $T^{-1}T = TT^{-1} = I$, and so $(T^{-1})^*$ is the inverse of T^* , giving $(1) \Leftrightarrow (2)$.

For $(1) \Rightarrow (3)$, we say that an operator F is bounded away from zero if

$$\|x\| \leq C\|F(x)\|$$

for some constant C . In our case:

$$\|x\| = \|T^{-1} \circ T(x)\| \leq \|T^{-1}\| \|T(x)\|$$

Since $(1) \Leftrightarrow (2)$, the same holds for T^* .

For $(3) \Rightarrow (4)$, We have $\|T(x)\| \geq \epsilon\|x\|$ and $\|T^*(x)\| \geq \epsilon\|x\|$ for some $\epsilon > 0$ and all $x \in \mathbb{H}$. Thus certainly both T and T^* are injective: if $T(x) = T(y)$ then

$$0 = \|T(x) - T(y)\| = \|T(x - y)\| \geq \epsilon\|x - y\|$$

hence $\|x - y\| = 0$, so $x = y$. Furthermore, $\|T(x) - T(y)\| \geq \|x - y\|$ implies that $T(\mathbb{H})$ is complete, hence a closed subspace of $\mathbb{H}$.

For $(4) \Rightarrow (5)$, by corollary 3.1.1 ($\overline{Y} = (Y^\perp)^\perp$) and proposition 3.2.1 we get

$$\overline{T(\mathbb{H})} = (T(\mathbb{H})^\perp)^\perp = (\ker(T^*))^\perp = \{0\}^\perp = \mathbb{H}$$

where the last equality come from T^* bein injective.

Finally, for $(5) \rightarrow (1)$, we use the open mapping theorem.

This allows us to generialize this result to the followin

Proposition 3.2.4: Inversing Maps

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be hilbert spaces and consider an operator $T \in \text{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$. Then there exists a unique operator $T^* \in B(\mathbb{H}_2, \mathbb{H}_1)$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

show that the map $T \rightarrow T^*$ is conjugate linear isometry of $\text{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$ to $\text{Hom}_k(\mathbb{H}_2, \mathbb{H}_1)$ and satisfies

$$\|T^*T\| = \|T^2\| = \|TT^*\|$$

Proof :

Let

$$\hat{T} = \begin{pmatrix} 0 & 0 \\ T & 0 \end{pmatrix} \in \text{End}_k(\mathbb{H}_1 \oplus \mathbb{H}_2)$$

By theorem 3.2.1, there is a unique function $\hat{T}^*$ such that

$$\langle \hat{T}(x_1, x_2), (y_1, y_2) \rangle = \langle (x_1, x_2), \hat{T}^*(y_1, y_2) \rangle$$

By definition of $\widehat{T}$, we see

$$\widehat{T}^* = \begin{pmatrix} 0 & T^* \\ 0 & 0 \end{pmatrix}$$

Hence we have that T^* is a function with domain and codomain $T^* : \mathbb{H}_2 \rightarrow \mathbb{H}_1$. It follows that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

and by theorem 3.2.1, T and T^* respect all the desired properties, completing the proof.

For the rest of this section, we will focus on $T \in B(\mathbb{H})$ instead of $T \in B(\mathbb{H}_1, \mathbb{H}_2)$. The reason for this is to find suitable decomposition theorems of T generalizing the notion of eigenvalues.

Recall that over finite dimensional vector spaces, $TT^* = T^*T$ implied T was diagonalizable, with the usual special case of $T = T^*$ (self-adjoint) being looked into. We would like to know if this notion of diagonalizability for normal operators remains true for infinite dimensional Hilbert spaces. We shall see that this is not the case, some more conditions are required (for ex. a special case of diagonalizable operators will be compact normal operators). Before we get into the technicalities, I want to put a word on how I think about $TT^* = T^*T$. In my mind, when two functions commute, it means that one function is somehow “invariant” with respect to the other. Think of PDE’s with equations that commute with the differential operator that makes it invariant to the Lorentzian metric. In this case, I like to think of T being somehow “symmetric”. Since $TT^*(x) = T^*T(x)$ for all $x \in \mathbb{H}$, and T^* is in some ways the dual (think of ref:HERE, the generalization), we see that T is “invariant” to the effect of the dual T^* . We shall see in a moment that the invariant to the dual is translated to T and T^* being metrically identical. In the simplest case of $T = T^*$, it is evident why this is the case. If $T = T^*$, we can deepen our intuition with complex numbers: if we consider $T : \mathbb{C} \rightarrow \mathbb{C}$ where $T(z) = z$, then $T^*(z) = \bar{z}$. Hence, $T = T^*$ if and only if $T(\mathbb{C}) \subseteq \mathbb{R}$, since only the real numbers are invariant under conjugation.

Definition 3.2.2: Normal Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *normal* if it commutes with its adjoint, i.e. if $T^*T = TT^*$

As an immediate consequence, we have

$$\|T(x)\| = \langle T^* \circ T(x), x \rangle^{1/2} = \langle T \circ T^*(x), x \rangle^{1/2} = \|T^*x\|$$

showing that T and T^* are metrically identical. In fact, by the parallelogram law (also known as polarization identity) if $T \in \text{End}_k(\mathbb{H})$ such that T and T^* are metrically identical, we can show $\langle T^* \circ T(x), y \rangle = \langle T \circ T^*(x), y \rangle$ for all x, y and so T is normal by lemma 3.2.1. By proposition 3.2.3, an operator is normal if and only if it is bounded away from zero.

Positive Operators

Definition 3.2.3: Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *positive* if $T = T^*$ and $\langle T(x), x \rangle \geq 0$ for all $x \in \mathbb{H}$. If $k = \mathbb{C}$, the positivity condition implies self-adjointness by the polarization identity.

Certainly, the sum of two positive operators is positive (geometrically, positive operators form a cone in $\text{End}_k(\mathbb{H})$). However the product is not positive, and need not be self-adjoint.

Example 3.5: Positive And Non-Positive Operators

here

However, if the operators commute (if the product is self-adjoint), we see that there is a product that is positive since $(ST = S^{1/2}TS^{1/2} \geq 0$ by what we'll show in the next proposition)

Let $B(\mathbb{H})_{sa} \subseteq \text{End}_{\mathbb{R}}(\mathbb{H})$ be the collection of self-adjoint operators. Then this is a closed subspace, this subspace intersected with the cone of positive operators defines an order $S \leq T$ if $T - S \geq 0$. By taking the case of 2 dimensional real subspaces and $\text{End}_{\mathbb{R}}(\mathbb{R}^2)$, this order is not total, not even a lattice order.

We require the following lemma to gain some important results (which will be an immediate consequence of the spectral theorem in ref:HERE)

Lemma 3.2.2: Square Root Lemma

If $S \leq T$ in $B(\mathbb{H})_{sa}$, then $A^*SA \leq A^*TA$ for every $A \in \text{End}_k(\mathbb{H})$. If moreover $S \leq T$, then $\|S\| \leq \|T\|$

Proof :

Since $S \leq T$, we know that $\langle (T - S)(y), y \rangle \geq 0$ for every $y \in \mathbb{H}$. Replacing y with $A(x)$ for appropriate $x \in \mathbb{H}$, it follows that $A^*(T - S)A \geq 0$, so $A^*SA \leq A^*TA$.

If $0 \leq S \leq T$, by the Cauchy-Schwarz inequality on the positive sesquilinear form $\langle S(\cdot), \cdot \rangle$, we get that for every pair of unit vectors $x, y \in \mathbb{H}$

$$|\langle S(x), y \rangle|^2 \leq \langle S(x), x \rangle \langle S(y), y \rangle \leq \langle T(x), x \rangle \langle T(y), y \rangle \leq \|T\|^2$$

Hence, $\|S\|^2 \leq \|T\|^2$ by lemma 3.2.1.

Lemma 3.2.3: Approximation Lemma

There is a sequence $(p_n) \subseteq \mathbb{R}[x]$ of polynomials with positive coefficients such that $\sum p_n$ converges uniformly on $[0, 1]$ to

$$t \mapsto 1 - (1 - t)^{1/2}$$

Proof :

This is immediate from real analysis, but we can also construct it explicitly. If $q_0 = 0$ and $q_n(t) = \frac{1}{2}(t + q_{n-1}(t)^2)$, then we can see that each coefficient of q_n is positive and that $q_n(t) \leq 1$ for every $t \in [0, 1]$. Moreover, since

$$2(q_{n+1} - q_n) = q_n^2 - q_{n-1}^2 = (q_n - q_{n-1})(q_n + q_{n-1})$$

it follows by induction that each polynomial $p_n = q_n - q_{n-1}$ has positive coefficients. Taking the q_n as elements of $C([0, 1])$, we see they converge pointwise to $[0, 1]$ to a function q such that

$2q(t) = t + q(t)^2$. Thus, $q(t) = 1 - (1 - t)^{1/2}$. Since $q \in C([0, 1])$ and $[0, 1]$ is compact, by the monotone convergence $q_n \nearrow q$ is in fact uniform by Dini's lemma, hence

$$\sum p_n = \lim q_n = q$$

completing the proof.

Proposition 3.2.5: Square-Rooting Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$ be a positive operator. Then there is a unique positive operator $T^{1/2}$ such that $(T^{1/2})^2 = T$. Moreover, $T^{1/2}$ commutes with every operator commuting with T .

Proof :

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Proposition 3.2.6: Invertible Positive Operator

A positive operator T is invertible in $\text{End}_k(\mathbb{H})$ if and only if $T \geq \epsilon I$ for some $\epsilon > 0$. In that case, $T^{-1} \geq$ and $T^{1/2}$ is invertible and $(T^{-1})^{1/2} = (T^{1/2})^{-1}$. If $T \leq S$, then $S^{-1} \leq T^{-1}$.

Proof :

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Projection

Let $W \subseteq \mathbb{H}$ be a closed subspace of a Hilbert space H . Then we know that $\mathbb{H} = W + W^\perp$. In particular, for each $y \in \mathbb{H}$, we have $y = w + w^\perp$, $w \in W$ and $w^\perp \in W^\perp$. Since

$$\alpha y_1 + \beta y_2 = (\alpha x_1 + \beta x_2) + (\alpha x_1^\perp + \beta x_2^\perp)$$

is the decomposition of a linear combination, it follows that $P : \mathbb{H} \rightarrow \mathbb{H}$ given by $P(y) = x$ is an operator in $\text{End}_k(\mathbb{H})$ with $\|P\| \leq 1$. Note P is idempotent ($P^2 = P$), self-adjoint, and positive since

$$\langle P(y_1), y_2 \rangle = \langle x_1, x_2 + x_2^\perp \rangle = \langle x_1, x_2 \rangle = \langle x_1 + x_1^\perp, x_2 \rangle = \langle y_1, P(y_2) \rangle$$

and

$$\langle P(y), y \rangle = \langle x, x + x^\perp \rangle = \|x\|^2 \geq 0$$

Definition 3.2.4: (Orthogonal) Projection

The map P in the above is called the *(orthogonal) projection*.

If P is a self-adjoint and idempotent in $\text{End}_k(\mathbb{H})$, then

$$W = \{x \in \mathbb{H} \mid P(x) = x\} = P(\mathbb{H})$$

is a closed subspace of $\mathbb{H}$. If $w^\perp \in W^\perp$, we have (since $P = P^*$) that

$$\|P(w^\perp)\|^2 = \langle w^\perp, P^2(w^\perp) \rangle = 0$$

Thus, P is an orthogonal projection of $\mathbb{H}$ onto W . Note that $I - P$ is the projection on $W^\perp$.

Projections are some of the easiest operators we may construct, and can be thought as the “characteristic function” from real analysis. The analogue of simple function is then obtained by decomposing $\mathbb{H}$ into a finite orthogonal sum

$$\mathbb{H} = W_1 \oplus W_2 \oplus \cdots \oplus W_n$$

corresponding to the projections $P_1, \dots, P_n$ which are pairwise orthogonal ($P_i P_j = 0$ for $i \neq j$ satisfying $\sum P_i = I$). We may then consider

$$T = \sum \lambda_n P_n$$

for scalars $\lambda_1, \lambda_2, \dots, \lambda_n \in k$. This may look familiar to the notion of *diagonalization*. Indeed, we may generalize the concept to infinite dimensions like so:

Definition 3.2.5: Diagonalizable

Let $T \in \text{End}_k(\mathbb{H})$ be an operator. Then T is said to be *diagonalizable* if there is an orthonormal basis $\{e_i \mid i \in I\}$ for $\mathbb{H}$ and a bounded set $\{\lambda_i \mid i \in I\} \subseteq k$ such that

$$T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$$

The number $\langle x, e_i \rangle$ are the coordinates for x in the basis $\{e_i\}$ and each λ_i is an *eigenvalue* for T corresponding to the eigenvector e_i . Denote P_i to be the projection of $\mathbb{H}$ onto ke_i , we have $P_i(x) = \langle x, e_i \rangle e_i$, so we may write the operator as:

$$T = \sum \lambda_i P_i$$

Note that the above sum is not necessarily convergent in the norm topology on $\text{End}_k(\mathbb{H})$, but only pointwise convergent (i.e. in convergent in the strong operator topology). If T is diagonalizable, then T^* is diagonalizable with

$$T^*(x) = \sum \overline{\lambda_i} \langle x, e_i \rangle e_i$$

with eigenvalues $\{\overline{\lambda_i}\}$. Furthermore, if T is diagonalizable, $T^*T = TT^*$ (with eigenvalues $|\lambda_i|^2$ with basis $\{e_i\}$), and hence T is normal. T is self-adjoint if and only if all λ_i are real and T is positive if and only if $\lambda_i \geq 0$ for all i . If $\mathbb{H}$ is finite-dimensional, every normal operator is diagonalizable. This is not generally true for infinite dimensions. In general, most “interesting” (normal) operators on infinite-dimensional Hilbert spaces are nondiagonalizable, and must be handled with a continuous analogue of the concept of a basis (this will be the spectrum, see section 3.3 and chapter 4).

Unitary Operators

Recall $T : \mathbb{H} \rightarrow \mathbb{H}$ is unitary if T is an isometric isomorphism from $\mathbb{H}$ onto itself (if $k = \mathbb{R}$ we say it is an *orthogonal operator*). By the polarization identity, we see that $U^*U = UU^* = I$, so that U is normal and invertible with $U^{-1} = U^*$. Conversely, if $U \in \text{End}_k(\mathbb{H})$ such that $U^{-1} = U^*$ then U is unitary since

$$\|U(x)\|^2 = \langle U^*U(x), x \rangle = \|x\|^2$$

So U is an isometry and is surjective since U is invertible. If $\{e_i \mid i \in I\}$ is an orthonormal basis for $\mathbb{H}$ and U is unitary, then $\{Ue_i \mid i \in I\}$ is a basis for $\mathbb{H}$. Conversely, we saw that every transition between orthonormal bases is given by a unitary operator. For future reference we denote the group of unitary operators $U(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ (it should be evident it forms a group).

Definition 3.2.6: Unitary Equivalent

Let $S, T \in \text{End}_k(\mathbb{H})$. Then we say that S and T are *unitarily equivalent* if $S = UTU^*$ for some unitary operator U .

It should be clear that unitary equivalent preserves norm, self-adjointness, normality, diagonalizability, and unitarity.

Definition 3.2.7: Partial Isometry

Let $U \in \text{End}_k(\mathbb{H})$. Then U is said to be a *partial isometry* if there is a closed subspace $X \subseteq \mathbb{H}$ such that $U|_X$ is isometric and $U|_{X^\perp} = 0$.

This immediately implies that $X^\perp = \ker(U)$, hence $X = \overline{U^*(\mathbb{H})}$. Setting $P = U^*U$, then

$$\langle P(x), x \rangle = \|U(x)\|^2 = \|x\|^2$$

for every $x \in X$, hence $P(x) = x$ by the Cauchy-Schwarz inequality. Furthermore, $P(x^\perp) = 0$ for every $x^\perp \in X^\perp$, so P is a projection of $\mathbb{H}$ onto X . Conversely, if $U \in \text{End}_k(\mathbb{H})$ such that $U^*U = P$ for some projection P , taking $X = P(\mathbb{H})$ and see from the equation $\|U(x)\|^2 = \langle P(x), x \rangle$ that U is isometric on X and 0 on $X^\perp$, and so U is a partial isometry. Using this, we can also show that U^* is a partial isometry. Since $U(I - P) = 0$,

$$(UU^*)^2 = UU^*UU^* = UPU^* = UU^*$$

so UU^* is self-adjoint idempotent, i.e. a projection. Note that U^* is the projection of $U(\mathbb{H})$ back onto $U^*(\mathbb{H})$ so that U^* is a *partial inverse* of U .

Be weary that there exist isometries that are not unitary (since they need not be surjective). For example, if $\mathbb{H}$ is a separable and has countable basis $\{e_n \mid n \in \mathbb{N}\}$, we have

$$S\left(\sum \alpha_n e_n\right) = \sum \alpha_n e_{n+1}$$

Then S (the unilateral shift operator) is an isometry of $\mathbb{H}$ onto the subspace $\{e_1\}^\perp$ and consequently S^* is a partial isometry of $\{e_1\}^\perp$ onto $\mathbb{H}$. In particular, $S^*S = I$, whereas SS^* is the projection onto $\{e_1\}^\perp$.

Theorem 3.2.2: Polar Decomposition of T

Let $T \in \text{End}_k(\mathbb{H})$. Then there is a unique positive operator $|T| \in \text{End}_k(\mathbb{H})$ such that

$$\|T(x)\| = \||T|(x)\|$$

and $|T| = (T^*T)^{1/2}$. Moreover, there is a unique partial isometry U with kernel $\ker(U) = \ker(T)$ and $U \circ |T| = T$. In particular

$$U^*U|T| = |T| \quad U^*T = |T| \quad UU^*T = T$$

Proof :

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We may think of U as the generalized “sign” of T and $|T|$ as the generalized “absolute value” of T . By the formula

$$T^* = |T|U^* = U^*(U|T|U^*)$$

and the unique of polar decomposition, we see that U^* is the sign of T^* , whereas $U|T|U^*$ is the absolute value of T^* . In general, the absolute value of a sum or product of noncommuting operators doesn't have much relation to the sum or product of the absolute values. Similarly, the sign (in the finite dimensional case), can't always be chosen to be unitary (the unilateral shift is an immediate counter example). For T to have the form $U|T|$ for some unitary operator U , it is necessary and sufficient that the closed subspaces $\ker(T)$ and $\ker(T^*)$ have the same dimension (finite or infinite). Easy cases of this general theorem are established in the next result

Proposition 3.2.7: Polar Decomposition For Isomorphisms

Let $T \in \text{End}_k(\mathbb{H})$ be invertible. Then the partial isometry in its polar decomposition is unitary

Proof :

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Proposition 3.2.8: Polar Decomposition For Normal Operators

Let $T \in \text{End}_k(\mathbb{H})$ be normal. Then there is a unitary operator W commuting with T, T^* , and $|T|$ such that $T = W|T|$

Proof :

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I want to add one more thing, I just don't know where: for any $T \in B(\mathbb{H})$, $T = T_1 + iT_2$ for self-adjoint operators T_1, T_2 . This lets us generalize the spectral theorem for normal operators, since we will know it for self-adjoint operator.

Lemma 3.2.4: lemma I

If $T = T^*$ and $\|T\| \leq 1$, the operator $U = T + i(I - T^2)^{1/2}$ is unitary and $T = \frac{1}{2}(U + U^*)$

Proof :

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Lemma 3.2.5: Lemma II

Let $S \in \text{End}_k(\mathbb{H})$ with $\|S\| < 1$. Then for every unitary U , there are unitaries U_1, V_1 such that

$$S + U = U_1 + V_1$$

Proof :

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Theorem 3.2.3: Russo-Dye-Gardner Theorem

Let $T \in \text{End}_k(\mathbb{H})$ with

$$\|T\| < 1 - \frac{2}{n} \quad n > 2$$

Then there are unitary operators $U_1, U_2, \dots, U_n$ such that

$$T = \frac{1}{n}(U_1 + U_2 + \dots + U_n)$$

Proof :

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Since the self-adjoint elements span $\text{End}_k(\mathbb{H})$, we see from lemma 3.2.4 that every $T \in \text{End}_k(\mathbb{H})$ is the linear combination of (at least 4) unitary operators. The estimate in the Russo-Dye-Gardner Theorem is much more precise with respect to the norm, and shows that every element in the open unit ball of $\text{End}_k(\mathbb{H})$ is a convex combination (indeed, a mean) of unitary operators. On the surface of the ball this need not be the case, in fact the unilateral shift cannot be expressed as a convex combination of unitaries (exercise 3.2.8 in Gert K. Pederson Analysis Now).

Proposition 3.2.9: Numerical Radius Of Operator

Let $\mathbb{H}$ be a complex hilbert space and $T \in \text{End}_{\mathbb{C}}(\mathbb{H})$. Then we deifn eth *numerical radius* of T as

$$|||T||| = \sup \{ \langle T(x), x \rangle \mid x \in \mathbb{H}, \|x\| \leq 1 \}$$

Then $\|\cdot\|$ is a norm on $\text{End}_{\mathbb{C}}(\mathbb{H})$ satisfying $\frac{1}{2}\|\cdot\| \leq |||\cdot||| \leq \|\cdot\|$. More over, $|||T^2||| \leq |||T|||^2$ for every T and $|||T||| = \|T\|$ if T is normal

Proof :

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3.3 Compact Operators

Throughout this section, let $\mathbb{H}$ be a hilbert space over $k \in \{\mathbb{R}, \mathbb{C}\}$ unless otherwise specified.

Definition 3.3.1: Finite Rank Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to have *finite rank* if $T(\mathbb{H})$ is finite-dimensional subspace of $\mathbb{H}$. Denote by $B_f(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ the set of all finite rank opeartors.

Clearly, $B_f(\mathbb{H})$ is a subspace, even a subalgebra, even an ideal of $\text{End}_k(\mathbb{H})$. If $T \in B_f(\mathbb{H})$, by proposition 3.2.1, we get an orthogonal decomposition $\mathbb{H} = T(\mathbb{H}) \oplus \ker(T^*)$, which shows that $T^*(\mathbb{H}) = T^*T(\mathbb{H})$, so T^* has finite rank. Thus, $B_f(\mathbb{H})$ is *self-adjoint* ideal in $\text{End}_k(\mathbb{H})$ ($(B_f(\mathbb{H}))^* = B_f(\mathbb{H})$ as a set). The subset $B_f(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ is similar in relation to $C_c(X) \subseteq C_b(X)$ when X is a locally compact Hausdorff space. In particular, these classes describe the local phenomena on $\mathbb{H}$ and on X . Passing to a limit in norm may destroy the “locality”. but enogh structure is preservd to make these “quasilocal” operators and functions very useful and interesting. We shall stud the closure of $B_f(\mathbb{H})$ in this section as a noncommutative analogue of $C_0(X)$ in function theory.

Lemma 3.3.1: Net And Compact Operators

There is a net $(P_\lambda)_{\lambda \in \Lambda}$ of projection in $B_f(\mathbb{H})$ such that $\|P_\lambda(x) - x\| \rightarrow 0$ for each $x \in \mathbb{H}$.

that is, the projection are “dense” in $B_f(\mathbb{H})$.

Proof :

We want to find a next P_n such that $\|P_n(x) - x\| \rightarrow 0$. Let $\{e_i \mid i \in I\}$ be an orthonormal basis for $\mathbb{H}$ and let N be a net of finite subsets of I ordered by inclusion. For each $n \in N$, let P_n denote the projection of $\mathbb{H}$ onto $\text{span}\{e_i \mid i \in n\}$ so that $(P_n)_{n \in N}$ is a net in $B_f(\mathbb{H})$.

Now, if $x \in \mathbb{H}$, then $x = \sum a_i e_i$, thus

$$\|P_n(x) - x\|^2 = \sum_{i \notin n} |a_i|^2$$

By Parseval’s identity, this tends to zero, completing the proof.

Theorem 3.3.1: Closed Unit Balls In Hilbert Spaces

Let $B \subseteq \mathbb{H}$ be a closed unit ball in a hilbert space $\mathbb{H}$. Then the following conditions on an operator $T \in \text{End}_k(\mathbb{H})$ are equivalent

1. $T \in \overline{B_f(\mathbb{H})}$
2. $T|_B$ is a weak-norm continuous function from B into $\mathbb{H}$
3. $T(B)$ is compact in $\mathbb{H}$
4. $\overline{T(B)}$ is compact in $\mathbb{H}$
5. each net in B has a subnet whose iamge under T converges in $\mathbb{H}$

Proof :

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Definition 3.3.2: Compact Operators

Let $T \in B(\mathbb{H})$. Then T is compact if $T(B)$ is compact. The set of compact operators is denoted by $B_0(\mathbb{H})$ to show that these operators “vanish at infinity”.

Lemma 3.3.2: Diagonalizable Operator Is Compact

Let $T \in \text{End}_k(\mathbb{H})$. Then T is comapct if and only if its eigenvalues $\{\lambda_i \mid i \in I\}$ belongs to $c_0(I)$

Proof :

Let $T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$. If $T \in B_0(\mathbb{H})$, then for any $\epsilon > 0$, define

$$I_\epsilon = \{i \in I \mid |\lambda_i| \geq \epsilon\}$$

If I_ϵ is infinite, the net $(e_i)_{i \in I_\epsilon}$ converges weakly to zero for any well-ordering of I_ϵ since $\langle e_i, x \rangle \rightarrow 0$ by Parsaval's identity. Sicne $\|T(e_i)\| = |\lambda_i| \geq \epsilon$ for $i \in I_\epsilon$ this contradicts theorem 3.3.1(3). Thus, I_ϵ must be finite for each $\epsilon > 0$, hence λ_i must vanish at infinity.

Conversely, if I_ϵ is finite for all $\epsilon > 0$. et

$$T_\epsilon = \sum_{i \in I_\epsilon} \lambda_i \langle \cdot, e_i \rangle e_i$$

Then T_ϵ has finite rank, and

$$\begin{aligned} \|(T - T_\epsilon)(x)\|^2 &= \left\| \sum_{i \notin I_\epsilon} \lambda_i \langle x, e_i \rangle e_i \right\|^2 \\ &= \sum_{i \notin I_\epsilon} |\lambda_i|^2 |\langle x, e_i \rangle|^2 \\ &\leq \epsilon^2 \|x\|^2 \end{aligned}$$

hence, $\|T - T_\epsilon\| \leq \epsilon$, thus $T \in B_0(\mathbb{H})$ by theorem 3.3.1(1), completing the proof.

Lemma 3.3.3: Eigen Vectors Between Adjoint

Let x be an eigenvector for a normal operator $T \in \text{End}_k(\mathbb{H})$ corresponding to an eigenvalue λ . Then x is an eigenvector for T^* corresponding to the eigenvalue $\bar{\lambda}$. Eigenvectors for T corresponding to different eigenvalues are orthogonal.

Proof :

Note that the operator $T - \lambda I$ is normal and its adjoint is $T^* - \bar{\lambda}I$. Conversely,

$$\|(T^* - \bar{\lambda}I)(x)\| = \|(T - \lambda I)(x)\| = 0$$

If $T(y) = \gamma y$ where $\gamma \neq \lambda$, then we may certainly assume $\lambda \neq 0$ so that

$$\begin{aligned} \langle x, y \rangle &= \lambda^{-1} \langle T(x), y \rangle \\ &= \lambda^{-1} \langle x, T^*(y) \rangle \\ &= \lambda^{-1} \langle x, \bar{\gamma}y \rangle \\ &= \lambda^{-1} \gamma \langle x, y \rangle \end{aligned}$$

Hence, $\langle x, y \rangle = 0$, showing they are different eigenvectors, completing the proof.

Lemma 3.3.4: Normal Compact Operator, then Eigenvalue Exists

Every normal, compact operator T on a complex Hilbert space $\mathbb{H}$ has an eigenvalue λ with $|\lambda| = \|T\|$

Proof :

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Theorem 3.3.2: Normal Compact, Then Diagonalizable

Every normal compact operator $T \in \text{End}_k(\mathbb{H})$ is diagonalizable and its eigenvalues (counted with multiplicity) vanish at infinite. Conversely, each such operator is normal and compact.

Proof :

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It will be convenient for future sections to introduce a new notation: Let $x \odot y$ represent the rank one operator in $\text{End}_k(\mathbb{H})$ determined by the vectors $x, y \in \mathbb{H}$ and the formula

$$(x \odot y)(z) = \langle z, y \rangle x$$

The map $(x, y) \mapsto x \odot y$ is sesquilinear map of $\mathbb{H} \times \mathbb{H}$ into $B_f(\mathbb{H})$. If $\|e\| = 1$ then $e \odot e$ is the one-dimensional projection of $\mathbb{H}$ on $\mathbb{C}e$. Every normal compact operator $\mathbb{H}$ can now be written by the above theorem in the form

$$T = \sum \lambda_i e_i \odot e_i$$

For suitable orthonormal basis $\{e_i \mid i \in I\}$. The sum will converge in norm, since either the set $I_0 = \{i \in I \mid \lambda_i \neq 0\}$ is finite (so that $T \in B_f(\mathbb{H})$) or else countably infinite in which case the sequence $\{\lambda_i \mid i \in I_0\}$ converges to zero.

Definition 3.3.3: Spectrum

We say that the compact set

$$\text{sp}(T) = \{\lambda_i \mid i \in I_0\} \cup \{0\}$$

is the *spectrum* of T .

For every continuous function f on $\text{sp}(T)$, define

$$f(T) = \sum f(\lambda_i) e_i \odot e_i$$

Then $f(T)$ is compact if and only if $f(0) = 0$ and the map $f \mapsto f(T)$ is an isometric $*$ -preserving homomorphism of $C(\text{sp}(T))$ into $\text{End}_k(\mathbb{H})$. If $f(z) = \sum \alpha_{nm} z^n \bar{z}^m$ is a polynomial of two commuting variables z and $\bar{z}$, then $f(T) = \sum \alpha_{nm} T^n (T^*)^m$. This result is the *spectral (mapping) theorem* for normal compact operators. In the next chapter, we shall generalize this to any normal operator.

Calkin Algebra and Index

Since $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$ is a closed ideal, we may define $B(\mathbb{H})/B_0(\mathbb{H})$ with the quotient norm¹. This quotient is given a name

Definition 3.3.4: Calkin Algebra

Let $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$. Then the algebra

$$B(\mathbb{H})/B_0(\mathbb{H})$$

is called the *Calkin Algebra* of $\mathbb{H}$

We shall see that many properties of $B(\mathbb{H})$ are conveniently expressed in terms of the Calkin algebra.

Definition 3.3.5: Compact Perturbation

If S and T are elements in $B(\mathbb{H})$ and $S - T \in B_0(\mathbb{H})$, we say that S is a *compact perturbation* of T .

This is another way of saying that S and T have the same image in the Calkin algebra. Such “local” perturbations happen frequently in application and properties of an operator that are invariant under

¹This is even a C^* -algebra

compact perturbations are highly used. The best known of these we shall study now is known as the *index*

Theorem 3.3.3: Atkinson's Theorem

Let $T \in B(\mathbb{H})$. Then the following are equivalent:

1. There is a unique operator $S \in B(\mathbb{H})$ such that ST and TS are the projections on $(\ker(T))^\perp$ and $\ker(T^*)^\perp$, respectively, and both projections have finite co-rank
2. For some operator $S \in B(\mathbb{H})$ both operators $ST - I$ and $TS - I$ are compact
3. the image of T is invertible in the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$
4. Both $\ker(T)$ and $\ker(T^*)$ are finite-dimensional subspaces and $T(\mathbb{H})$ is closed

Proof :

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In essence, Fredholm operators are operators that are almost invertible up to some finite-dimension being ignored, in particular they are invertible modulo compact operators.

Definition 3.3.6: Fredholm Operator

The operator satisfying the condition's in Atkinson's theorem are called *Fredholm operators*, and the class of them is denoted $F(\mathbb{H})$.

Definition 3.3.7: Index

For each $T \in F(\mathbb{H})$, define the *index* of T to be

$$\text{index}(T) = \dim(\ker(T)) - \dim(\ker(T^*))$$

If we choose S for T as in (1) of Atkinson's theorem and write $ST = I - P$ and $TS = I - Q$ with P and Q projection of rank, then another way of writing this would be

$$\text{index}(T) = \text{rank}(P) - \text{rank}(Q)$$

By (3) of Atkinson's Theorem, the product of Fredholm operators is again a Fredholm operator. In particular, every product $RT \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ and R is invertible in $B(\mathbb{H})$, and in this case

$$\text{index}(RT) = \text{index}(TR) = \text{index}(T)$$

since R is bijective. It should also be clear that $T^* \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ with $\text{index}(T^*) = -\text{index}(T)$. Finally, the partial inverse S to T in (1) of Atkinson's theorem is a Fredholm operator with $\text{index}(S) = -\text{index}(T)$. $\square$

Definition 3.3.8: Indexed Fredholm Operators

For each $n \in \mathbb{Z}$ define

$$F_n(\mathbb{H}) = \{T \in F(\mathbb{H}) \mid \text{index}(T) = n\}$$

All of these subsets are nonempty, since if S is the unilateral shift, then $S^n \in F_{-n}(\mathbb{H})$ and $(S^*)^n \in F_n(\mathbb{H})$ for every $n > 0$.

Lemma 3.3.5: Finite Rank And 0 Index

If $A \in B_f(\mathbb{H})$, then $I + A \in F_0(\mathbb{H})$

Proof :

Analysis now p.110

Corollary 3.3.1: Fredholm Alternative

If $A \in B_0(\mathbb{H})$ and $\lambda \in \mathbb{C} \setminus \{0\}$, then either $\lambda I - A$ is invertible in $B(\mathbb{H})$ or λ is an eigenvalue of A with finite multiplicity, in which case $\bar{\lambda}$ is an eigenvalue for A^* with the same multiplicity

Proof :

Let $T = \lambda I - A$. Since $T \in F_0(\mathbb{H})$, by lemma 3.3.5 we know that $T(\mathbb{H})$ is closed and that

$$\dim(\ker(T)) = \dim(\ker(T^*)) < \infty$$

If these dimension are 0, then $T(\mathbb{H}) = (\ker(T^*))^\perp = \mathbb{H}$ and $\ker(T) = \{0\}$, hence T is invertible by proposition 3.2.3, completing the proof.

This lemma implies that the spectrum of a compact operator (i.e. those λ such that $\lambda I - T$ is not invertible, as we'll see in the next chapter) consists of $\{0\}$ and the eigenvalues for T . IN particular, it is a countable subset of $\mathbb{C}$ with 0 as the only possible accumulation point.

Theorem 3.3.4: Index Invariant Under Compact Operator

Let $T \in F(\mathbb{H})$ and A be a compact operator. Then

$$\text{index}(T + A) = \text{index}(T)$$

Proof :

Analysis now p.111

Proposition 3.3.1: Fredholm Class Is Open

Each Fredholm class $F_n(\mathbb{H})$ for each $n \in \mathbb{H}$ is open in $B(\mathbb{H})$

Proof :

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Proposition 3.3.2: Grading Structure Of Fredholm Operators

If $T_1 \in F_n(\mathbb{H})$ and $T_2 \in F_m(\mathbb{H})$ then $T_1 T_2 \in F_{n+m}(\mathbb{H})$

Proof :

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We close with an interesting topological fact that brings some light to the index. Let G denote the group of invertible elements of the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$. and G_0 is the connected component of the identity (which is the group generated by the elements of the form $\exp(A)$ for some $A \in B(\mathbb{H})/B_0(\mathbb{H})$). Then G_0 is an open and closed subgroup of G and G/G_0 is a discrete group that labels the connected components in G . In this case, G/G_0 is isomorphic to $\mathbb{Z}$, and the index of a Fredholm operator T is the image of T under the composed quotient maps from $F(\mathbb{H})$ to G and from G to G/G_0 ($= \mathbb{Z}$)

3.4 Trace

4

Spectral Theory

In short, Spectral theory generalizes the notion of eigen vectors and values to infinite dimensional operators. In the previous chapter, we saw how there is a notion of diagonalizability for compact normal operators. In this chapter, we extend this result to normal operators. We also along the way learn some important dualities such as the Gelfand duality which has some useful applications, and is an interesting motivator for classical algebraic geometry.

4.1 Banach Algebras

Recall the definition of a Banach algebra

Definition 4.1.1: Banach Algebra Revisited

A *Banach Algebra* is an algebra A with a submultiplicative norm (so that it is continuous) such that the underlying vector space is a Banach space.

All the same applies for Banach Algebras as we expect. Note that for A/I to be a Banach algebra for an ideal I , I has to be closed, just like for subspaces. If A is unital and $A \neq \{0\}$, then it is certainly unique and $\|I\| \geq 1$ (why not =?? Analysis now p.128) If A has no units, we may try to isometrically embed it into a larger Banach Algebra $\hat{A}$ such that $A \subseteq \hat{A}$ becomes an ideal such that every nonzero ideal of $\hat{A}$ has a nonzero intersection with A (such an ideal is called an *essential ideal*). This can be thought as the algebraic counterpart of the notion of compactification of a topological space. For classical Banach algebra, there is usually a natural way of *adjoining a unit*, but there is always a general method of doing so that is the counter-part of one-point compactification. Take the direct sum $A \oplus k = \hat{A}$ along with the product

$$(x, \alpha)(y, \beta) = (xy + \alpha y + \beta x, \alpha\beta)$$

and the submultiplicative 1-norm. Then $A \oplus k$ is a unital Banach algebra with $I(0, 1)$ and contains A as an ideal of co-dimension 1.

Definition 4.1.2: Regular Representation

Let A be a Banach Algebra. Then the (left) *regular representation* $\rho : A \rightarrow B(A)$ by $\rho(x)(y) = xy$ for $x, y \in A$.

It should be verified that ρ is a norm decreasing algebra homomorphism, and if A is unital, the inequalities

$$\|A\| \leq \|\rho(A)\| \|I\| \leq \|A\| \|I\|$$

show that ρ is a homeomorphism. Renorming A by using the equivalent norm from $B(A)$, we may assume that $\|I\| = 1$ (if $A \neq \{0\}$), which we do.

Definition 4.1.3: Approximate Unit

A net $(E_\lambda)_{\lambda \in \Lambda}$ in the unit ball of a Banach algebra A is an *approximate unit* if

$$\lim E_\lambda x = \lim x E_\lambda = x$$

for all $x \in A$.

Example 4.1: Approximate Units

Can you think of approximate units for the non-unital Banach algebra $C_0(X)$, $B_0(\mathbb{H})$ and $L^1(\mathbb{R}^n)$? Show that the existence of an approximate unit guarantees that the regular representation ρ is an isometry

We shall denote $A^\times = \text{GL}(A)$ the set of all invertible elements of A (in algebra, we'd usually call these elements unital, but the term unit in functional analysis is more commonly reserved for the identity)

Lemma 4.1.1: Almost Invertible Elements

Let $x \in A$ where A is a unital Banach algebra with $\|x\| < 1$. Then $I - x \in \text{GL}(A)$ and

$$(I - x)^{-1} = \sum_{n=0}^{\infty} x^n$$

Proof :

Since the norm is submultiplicative $\|x^n\| \leq \|x\|^n$, thus the series $\sum x^n$ converges in A to an element y . Since $xy = yx = y - I$ by manipulating the above power series, it follows that $y = (I - x)^{-1}$, completing the proof.

Proposition 4.1.1: Properties Of $GL(A)$

Let A be a unital algebra and $GL(A)$ be it's multiplicative group. Then $GL(A)$ is an open subset of A , an the map $x \mapsto x^{-1}$ is a homeomorphism of $GL(A)$

Proof :

Let $x \in GL(A)$ and $y \in A$, Then by lemma 4.1.1

$$y = x - (x - y) = x(I - x^{-1}(x - y)) \in GL(A)$$

provided that $\|A^{-1}(A - B)\| < 1$. In particular, the ball $B_\epsilon(x)$ is contained in $GL(A)$ for $\epsilon < \|x^{-1}\|^{-1}$, which proves that $GL(A)$ is open. If $y \rightarrow a$, we see from the above equation and lemma 4.1.1 that $y^{-1} \mapsto a^{-1}$, thus inverting is a continuous process and therefore a homeomorphism since $(A^{-1})^{-1} = A$, completing the proof.

From ehre on out, all Banach algebra will be over $\mathbb{C}$, i.e. all Banach algebras will be *complex*. See exercise ref:HERE on the complexification of general real Banach algebras (or E.2.1.13 in Analysis Now).

Definition 4.1.4: Spectrum

Let A be a complex unital Banach algebra. Then for every $x \in A$, define the *spectrum* of x to be the set

$$\text{sp}(x) = \{\lambda \in \mathbb{C} \mid \lambda I - x \notin GL(A)\}$$

The complement of $\text{sp}(x)$ is the *resolvent set*.

On the resolvent set, we can define the resolvent function

$$R(x, \lambda) = (\lambda I - x)^{-1}$$

Definition 4.1.5: Spectral Radius

Let A be a complex unital Banach algebra, $x \in A$, and $\text{sp}(x)$ be its spectrum. Then the smallest $r \geq 0$ such that $\text{sp}(x) \subseteq B_r(0)$ is called the *spectral radius* of x , and is denoted $r(x)$. Thus

$$r(x) = \sup \{|\lambda| \mid \lambda \in \text{sp}(x)\}$$

Lemma 4.1.2: Build-Up Lemma I

Let $x \in A$ and $f(z) = \sum \alpha_n z^n$ be a holomorphic function in a region contained in the closed disk $B_r(0)$ with $\|x\| \leq r$. then we can define $f(x) = \sum \alpha_n x^n$ in A . Moreover, if $\lambda \in \text{sp}(x)$ and $|\lambda| \leq r$, then $f(\lambda) \in \text{sp}(f(x))$

Proof :

The series $f(x)$ is absolutely convergence ($\sum |\alpha_n| \|x\|^n < \infty$), so certainly $f(x) \in A$. Moreover

$$\begin{aligned} f(\lambda)I - f(x) &= \sum \alpha_n (\lambda^n I - x^n) \\ &= (\lambda I - x) \sum \alpha P_{n-1}(\lambda, x) \\ &= (\lambda I - x)y \end{aligned}$$

where $P_{n-1}(\lambda, x) = \sum_{k=0}^{n-1} \lambda^k A^{n-k-1}$ so that $\|P_{n-1}(\lambda, x)\| \leq nr^{n-1}$, which implies the series of polynomials converges in A to an element y commuting with x . We see from this computation that if $f(\lambda)I - f(x) \in \text{GL}(A)$ with inverse v , then yv will be the inverse of $\lambda I - x$, so tha $\lambda I - x \in \text{GL}(A)$

Lemma 4.1.3: Build-Up Lemma II

For every $x \in A$, we have

$$r(x) \leq \inf \|A^n\|^{1/n}$$

Proof :

If $|\lambda| > \|A\|$, thn by lemma 4.1.1

$$(\lambda I - x)^{-1} = \lambda^{-1}(I - \lambda^{-1}x)^{-1} = \sum \lambda^{-n-1}x^n$$

which shows that $r(x) \leq \|x\|$. Now if $\lambda \in \text{sp}(x)$, then $\lambda^n \in \text{sp}(x^n)$ by lemma 4.1.2. Thus $|\lambda^n| \leq \|x^n\|$ from what we've just shown. It follows that $r(x) \leq \|x^n\|^{1/n}$ for every n , completing the proof.

Theorem 4.1.1: Spectrum Is Compact

Let $x \in A$ be an element of a complex unital Banach algebra A . Then the spectrum of x , $\text{sp}(x)$ is a compact nonempty subset of $\mathbb{C}$, and the spectral radius of x is the limit of the convergence sequence $\|A^n\|^{1/n}$

Proof :

Analysis now p.131, really cool proof that I should take the time to write down!

A cool consequence of this is that we get a classical result of Representation theory known as Artin-Wedderburn for algebraically closed algebras:

Corollary 4.1.1: Algebraically Closed Division Algebra $\mathbb{C}$

Let A be a unital Banach algebra over $\mathbb{C}$. Then if A is a division ring, $A = \mathbb{C}$

Proof :

For each $x \in A$, there is some $\lambda \in sp(x)$ by the above theorem. Thus, $\lambda I - x \notin GL(A)$, hence $x = \lambda I$.

4.2 Continuous Functional Calculus

Let $T \in B(\mathbb{H})$ and consider $\sigma(T)$. Consider $B(\sigma(T))$, which is a $*$ -algebra where $*$ is complex conjugation. We may consider this algebra with the supremum norm. In this section, we'll be looking into various sub $*$ -algebra of $B(\sigma(T))$, for example the polynomials functions on $\sigma(T)$ or the continuous functions on $\sigma(T)$

Definition 4.2.1: Functional Calculus

Let $T \in B(\mathbb{H})$. Then a *functional calculus* of T is a continuous $*$ -homomorphism from $*$ -algebra of functions $(\sigma(T) \rightarrow \mathbb{C})$ to $B(\mathbb{H})$ which sends the constant function 1 to the identity operator, and sends the identity function $x \mapsto x$ to T

Since it's a $*$ -homomorphism, we have

$$\varphi(\bar{f}) = \varphi(f)^*$$

since we have the supremum norm and the $*$ -homomorphism is continuous, $f_n \rightarrow f$ in the supremum norm then $f_n(T) \rightarrow f(T)$ in the norm topology of $B(\mathbb{H})$.

(more here, this is really cool!)

4.3 Gelfand Duality

A

Important Intuition

This is a conversation I had with Gemini that I really want to incorporate somewhere as it gave some good insights. I will save it here for now:

Zak Katz: Let $F(x_1, \dots, x_n, t)$ represent a PDE w.r.t time. Let's say $u(x, t)$ represent a solution and $f_t(x) = u(x, t)$ is a “state” at time t . Let S_t be the “time-shift” operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a .

My question is: Is S_t always self-adjoint? If yes, can you prove it, and if not, can you give a counter-example?

Raycast AI:

Tool: `web_search`

No, the time-shift operator S_t is not always self-adjoint.

In fact, for many common PDEs (like the wave equation, transport equation, or Schrödinger equation), S_t is almost never self-adjoint. It is usually unitary (energy-conserving) or a contraction semi-group (dissipative).

Counter-Example: The Linear Transport Equation

Consider the simple advection equation in one dimension:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

where $c \neq 0$ is a real constant velocity.

1. **The Operator:** The solution is a translation of the initial state $f(x)$:

$$u(x, t) = f(x - ct)$$

Therefore, the time-shift operator S_t acts as:

$$(S_t f)(x) = f(x - ct)$$

2. **The Adjoint:** To check if S_t is self-adjoint, we compute its adjoint S_t^* using the standard L^2 inner product $\langle f, g \rangle = \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx$.

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(x - ct) \overline{g(x)} dx$$

Perform a change of variables $y = x - ct$, so $x = y + ct$ and $dx = dy$:

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(y) \overline{g(y + ct)} dy$$

$$\langle S_t f, g \rangle = \langle f, S_{-t} g \rangle$$

This shows that the adjoint is the inverse shift:

$$S_t^* = S_{-t}$$

3. **Conclusion:** For S_t to be self-adjoint, we require $S_t = S_t^*$. This would imply $S_t = S_{-t}$, or $f(x - ct) = f(x + ct)$ for all functions f . This is clearly false for $c \neq 0$ and $t \neq 0$. Thus, S_t is **not self-adjoint**.

General Context

- **Unitary vs. Self-Adjoint:** In conservative systems (like Quantum Mechanics or the Wave Equation), the evolution operator S_t is typically **unitary**, meaning $S_t^* = S_t^{-1} = S_{-t}$. A unitary operator is rarely self-adjoint (unless $S_t^2 = I$, which is not true for general time evolution).
- **When is it Self-Adjoint?** S_t is self-adjoint in specific dissipative systems, such as the **Heat Equation** ($u_t = \Delta u$). In this case, $S_t = e^{t\Delta}$. Since the Laplacian Δ is a self-adjoint operator (with appropriate boundary conditions), the operator exponential $e^{t\Delta}$ is also self-adjoint for real t .

Zak Katz: I see. Does that mean that S_t is not always “diagonalizable”?

Raycast AI: No, that is a common misconception! **“Not self-adjoint” does not mean “not diagonalizable.”**

The time-shift operator S_t for the Transport Equation (which I used as a counter-example previously) is diagonalizable, even though it is not self-adjoint.

Here is the breakdown of why this works and when it fails.

1. The Key Concept: Normal Operators

In linear algebra and functional analysis, a deeper property governs diagonalizability: **Normality**.

An operator A is **Normal** if it commutes with its adjoint:

$$AA^* = A^*A$$

The **Spectral Theorem** states that **any Normal operator is diagonalizable** (specifically, unitarily diagonalizable).

- **Self-Adjoint Operators** are Normal ($A = A^*$, so $AA = AA$).
- **Unitary Operators** are Normal ($A^* = A^{-1}$, so $AA^{-1} = A^{-1}A = I$).

2. Revisiting the Transport Equation (The “Yes” Case)

Let’s look at the Transport Equation again ($u_t + u_x = 0$), where S_t shifts the function: $f(x) \rightarrow f(x-t)$.

- We found that $S_t^* = S_{-t}$ (the inverse shift).
- Does it commute?

$$S_t S_t^* = S_t S_{-t} = S_0 = I$$

$$S_t^* S_t = S_{-t} S_t = S_0 = I$$

- Because $S_t S_t^* = S_t^* S_t$, the operator is **Normal**.

How do we “diagonalize” it? We use the **Fourier Transform**. In the frequency domain (Fourier space), the shift operator becomes multiplication:

$$\mathcal{F}(S_t f)(k) = e^{-ikt} \hat{f}(k)$$

Multiplication by a scalar (or function) is the infinite-dimensional equivalent of a diagonal matrix. The Fourier basis functions e^{ikx} are the “eigenvectors” of the time-shift operator.

Conclusion: The Transport operator is not self-adjoint, but it is normal, so it **is** diagonalizable.

3. When is S_t NOT Diagonalizable? (The “No” Case)

For S_t to fail to be diagonalizable (to be “defective”), the system must be **Non-Normal** in a specific way that creates “Jordan Blocks.” This happens when eigenvectors coalesce and you lose a complete basis (this is called an **Exceptional Point** in physics).

Example: A “Defective” System of PDEs Consider a coupled system of two variables $u(x, t)$ and $v(x, t)$:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

Here, the change in u depends on v , but v is constant in time.

1. **Solve it:**

- $v(x, t) = v(x, 0)$ (constant)
- $u_t = v(x, 0) \implies u(x, t) = u(x, 0) + t \cdot v(x, 0)$

2. **The Time-Evolution Operator S_t :** We can write S_t as a matrix acting on the vector $\begin{pmatrix} f \\ g \end{pmatrix}$:

$$S_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Is it Diagonalizable?** Check the matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.

- The eigenvalues are $\lambda = 1$ (with algebraic multiplicity 2).
- However, there is only **one** independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
- Because we are missing a second eigenvector, this operator is **not diagonalizable** for any $t \neq 0$.

Summary Table

Type of PDE	Operator Property	Self-Adjoint?	Diagonalizable?
Heat Eq. ($u_t = \Delta u$)	Self-Adjoint	Yes	Yes (Real eigenvalues)
Transport / Wave	Unitary	No	Yes (Complex eigenvalues)
Defective System	Non-Normal	No	No (Jordan Block)

Bottom Line: Most standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because they are typically governed by Normal operators. You usually have to construct specific coupled systems or non-Hermitian boundary conditions to break diagonalizability.

Zak Katz: Let me be clear: S_t shifts *in time* not in space. So to be clear: let a PDE $F(x_1, \dots, x_n, t) = 0$ PDE w.r.t time t , if $u(x, t)$ is a solution and $f_t(x) = u(x, t)$ is a “state” at time t , and Let S_t be the “time-shift” operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a . (NOT the space shift operator $f(x) \rightarrow f(x+t)$). Is S_t always diagonalizable?

Raycast AI: Thank you for the clarification. I understand the distinction you are making: you are asking about the **time-evolution operator** (often called the Propagator, $U(t)$ or S_t) that maps the initial state $u(\cdot, 0)$ to the state at time t , $u(\cdot, t)$.

The answer remains: **No, S_t is not always diagonalizable.**

While many standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because their generators are Normal (Self-Adjoint or Skew-Adjoint), it is very easy to construct physical systems where S_t is **defective** (non-diagonalizable).

This happens specifically when the system exhibits **Linear Instability** or **Critical Damping** (often associated with “Exceptional Points” in modern physics).

The Mathematical Reason: Jordan Blocks

If a PDE can be written in the form $\frac{\partial}{\partial t}\psi = L\psi$, the time-evolution operator is formally $S_t = e^{Lt}$.

For S_t to be diagonalizable, the generator L must possess a complete basis of eigenfunctions. If L has a **Jordan Block** (i.e., it has generalized eigenvectors but not enough true eigenvectors), then the exponential $S_t = e^{Lt}$ will also possess a Jordan block structure for any $t \neq 0$, making it non-diagonalizable.

Here are two concrete counter-examples.

Counter-Example 1: The “Resonant” System (Algebraic Growth)

Consider a system of two fields, $u(x, t)$ and $v(x, t)$, on a domain Ω . The PDE is:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

where λ is a constant. This is a standard linear PDE system.

1. **Find the Solution:** This is a coupled system.

- $v_t = \lambda v \implies v(x, t) = e^{\lambda t}v(x, 0)$.
- $u_t = \lambda u + v$. Substituting the solution for v : $u_t = \lambda u + e^{\lambda t}v(x, 0)$.
- The solution for u involves a secular term (a linear t):

$$u(x, t) = e^{\lambda t}u(x, 0) + te^{\lambda t}v(x, 0)$$

2. **The Time-Evolution Operator S_t :** Writing this as an operator matrix acting on the initial vector $\psi_0 = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$:

$$S_t = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Why it is NOT Diagonalizable:** Look at the matrix part $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.

- **Eigenvalues:** It has a single eigenvalue $\mu = 1$ (repeated).
- **Eigenvectors:** It has only **one** linearly independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
- Because we need two independent eigenvectors to span the 2D space of fields at every point x , and we only have one, S_t is **not diagonalizable** for any $t \neq 0$.

Physical Interpretation: The term $te^{\lambda t}$ represents “algebraic growth” (or transient growth). Whenever you see a solution of the form $t \times (\text{exponential})$, it is the signature of a non-diagonalizable operator. A diagonalizable operator would only have pure sums of exponentials.

Counter-Example 2: Critical Damping (Classical Mechanics / PDE)

Consider the Damped Harmonic Oscillator equation for a field $u(x, t)$, which is a second-order PDE in time:

$$u_{tt} + 2\gamma u_t + \gamma^2 u = 0$$

(Here $\gamma > 0$ is the damping coefficient).

To analyze S_t , we convert this to a first-order system (state-space form) by defining $v = u_t$:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

The Critical Case: The eigenvalues of the matrix $L = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}$ are found by $\det(L - \omega I) = 0$.

$$\omega^2 + 2\gamma\omega + \gamma^2 = (\omega + \gamma)^2 = 0$$

We have a repeated eigenvalue $\omega = -\gamma$.

Because the matrix L is not the identity matrix but has a repeated eigenvalue, it is a non-trivial Jordan block. The solution is:

$$u(t) = C_1 e^{-\gamma t} + C_2 t e^{-\gamma t}$$

The presence of the $t e^{-\gamma t}$ term confirms that the propagator S_t cannot be diagonalized. If it were diagonalizable, we could write the solution purely as $u(t) = A e^{\omega_1 t} + B e^{\omega_2 t}$, which implies strictly exponential decay without the linear polynomial pre-factor.

Summary

- **Is S_t always diagonalizable? No.**
- **When does it fail?** It fails at **Exceptional Points**, where two or more eigenmodes of the system coalesce (become parallel), making the set of eigenvectors incomplete. This results in solutions that grow or decay with polynomial factors (like t , t^2) multiplying the exponentials.

Zak Katz: Okay. Now, when it *is* diagonalizable, is it *always* the case that we can use Fourier transforms to find an eigenbasis?

Raycast AI: No. Absolutely not.

Just because an operator is diagonalizable does not mean the Fourier Transform provides the basis.

The **Fourier Transform** is a very specific tool that only diagonalizes operators that are **Translation Invariant** (i.e., operators with **Constant Coefficients** defined on the entire space $\mathbb{R}^n$).

If your PDE has **variable coefficients** or is defined on a **bounded domain**, the Fourier basis (e^{ikx}) will generally fail to diagonalize the operator, yet a different eigenbasis might still exist perfectly fine.

Here is the hierarchy of logic:

1. **Diagonalizable:** Means *there exists* a basis $\{\varphi_n(x)\}$ such that the operator acts as multiplication on these functions.
2. **Fourier Diagonalizable:** Means that specific basis is $\{\varphi_k(x) = e^{ikx}\}$.

When does Fourier work?

Fourier analysis works **only** if your operator L commutes with the translation operator T_a (where $T_a f(x) = f(x + a)$).

- **Example:** The Heat Equation $u_t = u_{xx}$.
 - The second derivative behaves the same at $x = 0$ as it does at $x = 100$. Space is homogeneous.
 - Therefore, the eigenfunctions are plane waves e^{ikx} .

When does Fourier fail (but the operator is still diagonalizable)?

Fourier fails whenever **spatial symmetry is broken**. This happens in three common scenarios:

1. **Variable Coefficients (Space is not homogeneous)** Consider the **Quantum Harmonic Oscillator** (one of the most famous diagonalizable systems in physics).

$$i \frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + x^2 \psi$$

- **Is it Diagonalizable?** Yes. The Hamiltonian is Self-Adjoint. It has a discrete, perfect set of eigenfunctions.
- **Are they Fourier Modes?** No. The term x^2 breaks translation invariance (physics looks different at $x = 0$ vs $x = 100$).
- **The Actual Basis:** The eigenfunctions are **Hermite Functions** (a Gaussian multiplied by a polynomial):

$$\varphi_n(x) = H_n(x) e^{-x^2/2}$$

If you tried to apply a Fourier Transform to this equation, the x^2 multiplication term turns into a Laplacian in k -space ($-\partial_k^2$), so you effectively just swap the PDE for an identical PDE in frequency space without solving it.

2. **Bounded Domains (The “Boundary Condition” Problem)** Consider the heat equation $u_t = u_{xx}$ on a finite interval $x \in [0, L]$ with Dirichlet boundary conditions $u(0) = u(L) = 0$.

- **Is it Diagonalizable?** Yes.
- **Is it the Fourier Transform?** No. The Continuous Fourier Transform uses e^{ikx} for any real k .

- **The Actual Basis:** The basis is strictly **Sine functions**: $\sin(\frac{n\pi x}{L})$.
 - While related to Fourier *Series*, the Continuous Fourier *Transform* basis (e^{ikx}) is invalid here because e^{ikx} does not satisfy the boundary conditions (it doesn't vanish at 0 and L). You are forced onto a discrete subset of the frequency spectrum.

3. Different Geometries If your PDE is on a sphere (like weather patterns on Earth):

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

- **Is it Diagonalizable?** Yes.
- **The Actual Basis:** The eigenfunctions are **Spherical Harmonics** $Y_l^m(\theta, \varphi)$.
- Standard planar Fourier modes (e^{ikx}) make no sense here.

Summary Table

PDE Feature	Is S_t Diag.?	Fourier?	The Actual Eigenbasis
Constant Coeffs on $\mathbb{R}$ ($u_t = u_{xx}$)	Yes	Yes	Plane Waves (e^{ikx})
Variable Coeffs ($u_t = xu_x$)	Yes	No	Power functions (x^λ)
Potential Well ($u_t = i\Delta u + iV(x)u$)	Yes	No	Hermite / Legendre / Bessel
Bounded Domain ($x \in [0, 1]$)	Yes	No	Fourier <i>Series</i> (Sines/Cosines)

Conclusion: Diagonalizability guarantees that *some* coordinate system exists where the operator is simple. It does not guarantee that the coordinate system is the one given by Fourier.

Zak Katz: Okay. Is it a necessary and sufficient condition that S_t is diagonalizable by the Fourier basis iff the diagonalizable operator is translation invariant? Just to make sure I understand what translation invariant means here, we are talking about *space* translation, and we mean that if S_t is our operator and τ_a is our space-translation, then:

$$S_t(\tau_a(f)) = \tau_a(S_t(f))$$

Raycast AI: Yes, precisely.

For a linear operator S_t acting on functions defined on the entire space $\mathbb{R}^n$ (or a periodic domain), the condition is **necessary and sufficient**:

Theorem: A bounded linear operator S_t is diagonalized by the Fourier Transform if and only if it is **Translation Invariant**.

Your definition of translation invariance is exactly correct:

$$[S_t, \tau_a] = S_t \tau_a - \tau_a S_t = 0 \quad \text{for all } a \in \mathbb{R}^n$$

where τ_a is the spatial shift operator.

Here is the logic for why this is an “if and only if” relationship.

1. Why is it Sufficient? (Translation Invariant $\implies$ Fourier Diagonal)

This relies on a fundamental principle of linear algebra: **Commuting operators share a common eigenbasis.**

1. Consider the family of translation operators $\{\tau_a\}_{a \in \mathbb{R}}$.
2. What are the eigenfunctions of a translation? Let's verify that the plane waves e^{ikx} are eigenfunctions of τ_a :

$$\tau_a(e^{ikx}) = e^{ik(x+a)} = e^{ika} \cdot e^{ikx}$$

The eigenvalue is the constant number e^{ika} , and the eigenfunction is e^{ikx} .

3. Since your time-evolution operator S_t commutes with **every** translation operator τ_a (for all shifts a), it must share the same set of eigenfunctions.
4. Therefore, the eigenfunctions of S_t must also be e^{ikx} .
5. If the eigenfunctions are e^{ikx} , then by definition, the operator is diagonalized by the Fourier Transform.

2. Why is it Necessary? (Fourier Diagonal $\implies$ Translation Invariant)

If an operator is diagonal in Fourier space, it acts as a simple multiplication. Let's call the Fourier transform $\mathcal{F}$.

1. Assume S_t is a Fourier Multiplier: $\widehat{S_t u}(k) = m(k) \hat{u}(k)$ for some multiplier function $m(k)$.
2. The translation operator τ_a in Fourier space is also just a multiplication (by the phase factor e^{ika}).
3. Since regular multiplication of scalars is commutative ($a \cdot b = b \cdot a$), the multiplication by $m(k)$ commutes with the multiplication by e^{ika} .
4. Therefore, S_t and τ_a commute.

Summary

This concept is the bedrock of signal processing and linear systems theory (where these are called **Linear Time-Invariant (LTI)** systems, or in your PDE context, “Linear Space-Invariant” systems).

- Does S_t commute with space shifts?

- **Yes:** You can use Fourier Transforms (e.g., Heat Eq, Wave Eq, Schrödinger with constant potential).
- **No:** You cannot use Fourier Transforms to find the eigenbasis (e.g., Schrödinger with Harmonic Oscillator potential, variable coefficient Wave Eq).

Zak Katz: Right. Is there a “theory” for diagonalizable operators S_t ?

Raycast AI: Yes, there is a vast and rich “theory” for this, but it doesn’t have a single name like “The Theory of Diagonalizable Operators.” Instead, it is split into different fields of **Functional Analysis** depending on *why* the operator is diagonalizable.

If you are looking for the generalized version of “Fourier Analysis” for operators that are **not** translation invariant (i.e., not solvable by Fourier Transform), here is the map of the theory.

1. The “Gold Standard”: The Spectral Theorem

This is the master theory. It applies if your operator S_t (or its generator L) is **Normal** ($LL^* = L^*L$).

- **The Theory: Spectral Theory of Normal Operators.**
- **What it says:** Any normal operator on a Hilbert space is unitarily equivalent to a multiplication operator.
- **Physical Meaning:** There *always* exists an orthonormal basis of eigenfunctions (or a “continuous basis” / spectral measure) that diagonalizes the operator.
- **How to find the basis:** This is the hard part. It usually requires solving the **Sturm-Liouville Problem** (for 1D) or using **Scattering Theory** (for unbounded domains).

2. The “Variable Coefficient” Theory: Sturm-Liouville Theory

If your PDE has variable coefficients but is in 1D (or separable), this is the standard replacement for Fourier analysis.

- **The Setup:** Your generator looks like $L = \frac{d}{dx}[p(x)\frac{d}{dx}] + q(x)$.
- **The Result:** Just like Fourier series uses $\sin(nx)$ and $\cos(nx)$, Sturm-Liouville theory guarantees a set of orthogonal eigenfunctions $\varphi_n(x)$ weighted by a density function.
- **Example:** Legendre polynomials, Bessel functions, Hermite polynomials (Quantum Harmonic Oscillator). These are the “custom Fourier bases” for specific geometries or potentials.

3. The “Abstract” Theory: Semigroup Theory (C_0 -Semigroups)

This is the modern mathematical framework for time-evolution operators S_t .

- **The Setup:** You treat S_t as a “one-parameter semigroup” $T(t)$ acting on a Banach space.
- **The Theorem:** The **Hille-Yosida Theorem**.
- **What it does:** It characterizes exactly when a differential operator L generates a stable time-evolution S_t . While Hille-Yosida doesn’t guarantee *diagonalizability* (it only guarantees existence and uniqueness of the flow), it is the starting point for analyzing the spectrum of S_t .
- **Spectral Mapping Theorem:** This theorem relates the eigenvalues of the generator L to the eigenvalues of the propagator S_t (basically $\sigma(S_t) = e^{t\sigma(L)}$). If you know the spectrum of L , you know the spectrum of S_t .

4. The “Non-Normal” Theory: Riesz Bases and Spectral Operators

What if S_t is diagonalizable but the eigenvectors are **not orthogonal**? (This happens in non-conservative systems, like convection-diffusion with non-standard boundary conditions).

- **The Theory: Non-Self-Adjoint Spectral Theory** (or the theory of **Riesz Bases**).
- **What it says:** You can still expand solutions in a basis of eigenfunctions, but because the basis is not orthogonal ($\langle \varphi_n, \varphi_m \rangle \neq \delta_{nm}$), you need a “dual basis” $\{\psi_n\}$ to find the coefficients.
 - Expansion: $f = \sum a_n \varphi_n$
 - Coefficients: $a_n = \langle f, \psi_n \rangle$ (instead of $\langle f, \varphi_n \rangle$).
- **Why it matters:** This is currently a very hot topic in fluid dynamics and laser physics, where systems are often “almost” normal but mathematically non-self-adjoint.

Summary: Which “Theory” do you need?

If your operator S_t is...	The Theory is called...	The Basis is...
Translation Invariant	Harmonic Analysis	Fourier Transform
Self-Adjoint / Normal	Spectral Theory	Orthonormal Eigenbasis
1D Variable Coeffs	Sturm-Liouville Theory	Special Functions
Not Normal (but diag.)	Non-Self-Adjoint Theory	Riesz Basis (Biorthogonal)
Not Diagonalizable	Jordan Form	Generalized Eigenvectors